The core architectural components of Azure:

What is Microsoft Azure?
Microsoft Azure is a cloud computing platform that provides cloud services for:

- Compute
- Networking
- Storage
- Databases
- Security
- AI/ML
- Analytics

through Microsoft's global cloud infrastructure.

❓ **Why is it Important?**
Azure helps organizations:
✓ Build applications
✓ Deploy cloud infrastructure
✓ Store/manage data
✓ Scale resources globally
✓ Use cloud services without managing physical hardware

👉 **Azure = Microsoft's global cloud platform for building and managing IT services**

☁️ **What Azure Offers**
Azure provides:
✓ Compute services
✓ Storage solutions
✓ Networking infrastructure
✓ Databases
✓ AI & ML services
✓ Security services
✓ DevOps & monitoring tools

🌟 **Core Azure Service Categories**

☁️ 1️⃣ **Compute**
Purpose:
Provides processing power/resources.

Examples

- Virtual Machines (VMs)
- App Services
- Azure Functions
- Kubernetes

👉 **Compute = Running applications/workloads**

🌐 2️⃣ **Networking**
Purpose
Connect cloud resources securely.

Examples

- Virtual Network (VNet)
- Load Balancer
- VPN Gateway
- DNS

👉 **Networking = Communication/connectivity**

💾 3️⃣ **Storage**
Purpose
Store files/data/backups.

Examples
- Blob Storage
- File Storage
- Queue Storage
- Managed Disks

👉 **Storage = Saving data in cloud**

🗄️ 4️⃣ **Databases**
Purpose
Managed database services.

Examples
- Azure SQL
- Cosmos DB
- PostgreSQL
- MySQL

👉 Databases = Structured data management

🤖 5️⃣ **AI + ML**
Purpose
Artificial Intelligence & Machine Learning services.

Examples
- Azure OpenAI
- AI Services
- Machine Learning

👉 AI/ML = Intelligent cloud services

🔐 6️⃣ **Identity & Security**
Purpose
Protect users/resources.

Examples
- Microsoft Entra ID
- Key Vault
- Defender for Cloud
- Firewall

👉 **Security = Identity + Protection**

⚙️ 7️⃣ **DevOps + Management**
Purpose
Deployment, monitoring, automation, governance.

Examples
- Azure DevOps
- Azure Monitor
- ARM Templates
- Azure Policy

👉 **DevOps/Management = Automation & monitoring**

🌍 8️⃣ **IoT**
Purpose
Connect/manage smart devices.

Examples
- IoT Hub
- IoT Central

👉 **IoT = Internet-connected devices**

📊 9️⃣ **Analytics**
Purpose
Data analysis & insights.

Examples
- Synapse Analytics
- Data Factory
- Databricks

👉 **Analytics = Data processing & insights**

🔄 🔟 **Integration**
Purpose
Connect applications/services together.

Examples
- Logic Apps
- API Management
- Event Grid

👉 **Integration = Connecting systems/services**

Azure is mainly:
👉 A global cloud computing platform

Which Azure category provides VMs?
👉 Compute

Which Azure category provides VNets?
👉 Networking

Which Azure category provides Azure SQL?
👉 Databases

Which Azure category provides Blob Storage?
👉 Storage

❤️**Example:**
Company hosts cloud application in Azure
Uses:
✓ VM (Compute)
✓ VNet (Networking)
✓ Azure SQL (Database)
✓ Blob Storage (Storage)
✓ Azure Monitor (Monitoring)
All together create cloud infrastructure.

Get Started with Azure Accounts:

What is an Azure Account?
An Azure account is the identity used to access Azure services and subscriptions.

❓ **Why is it Important?**
Azure account allows users to:
✓ Access Azure Portal
✓ Create subscriptions
✓ Deploy resources
✓ Manage cloud services

👉 **Azure Account = Entry point to Azure cloud**

🧠 **IMPORTANT UNDERSTANDING**

1️⃣ **Azure Account:** Identity/account used to access Azure.

Example
Your student Azure account.

👉 **Azure Account = Login identity**

2️⃣ **Subscription:** Logical unit for:

- Billing
- Access control
- Resource management

Why important?
Resources are created inside subscriptions.

Example

- Dev/Test subscription
- Production subscription

👉 **Subscription = Billing & management boundary**

3️⃣ **Resource Group:** Logical container for Azure resources.
Why important?
Organizes related resources together.

Example
Resource group containing:

- VM
- Database
- Storage
- Networking

for one application/project.

👉 **Resource Group = Folder for resources**

4️⃣ **Resources:** Actual Azure services/components.

Examples

- Virtual Machine
- Storage Account
- VNet
- SQL Database

👉 **Resources = Actual cloud services**

🌟 Azure Free Account
Includes
✓ Free Azure services
✓ Limited free credits
✓ Access to always-free services

🎓 Azure Student Account
Benefits
✓ No credit card required
✓ Free Azure credits
✓ Free developer tools
✓ Learning/practice environment

⚠️ **VERY IMPORTANT PRACTICAL RULE**
👉 **Always monitor resource usage**
Because some resources may consume credits if left running.

Which level handles billing/access management?
👉 Subscription

Where are Azure resources organized?
👉 Resource Groups

Every Azure resource belongs to?
👉 One Resource Group

Azure hierarchy order?
👉 Account → Subscription → Resource Group → Resource

Azure Physical Infrastructure: Azure's physical infrastructure consists of:

- Datacenters
- Availability Zones
- Regions
- Geographies

These work together to provide high availability, reliability, and disaster recovery.

👉 **Azure organizes datacenters into regions and availability zones for reliability and global availability**.

🌎 **Azure Geography:** A geographical area containing one or more Azure regions.

Examples

- United States
- Europe
- Asia

Purpose

- Data residency
- Compliance requirements
- Regulatory boundaries

👉 **Geography = Large area containing multiple regions**

🌎 **Azure Region:** A region is a collection of one or more datacenters located in a specific geographic area.

Examples

- East US
- West US
- Central India
- South India

Purpose

- Resource deployment
- Low latency
- High availability

Important
When creating Azure resources, you usually select a region.

👉 **Region = Location where resources are deployed**

🏢 **Datacenter:** Physical facility containing:

- Servers
- Storage
- Networking equipment
- Power & cooling systems

Purpose
Runs Azure services physically.

👉 **Datacenter = Physical building containing cloud hardware**

🛡️ **Availability Zones (VERY IMPORTANT):** Physically separate datacenters within the same Azure region.
Each zone has:
✓ Independent power
✓ Independent cooling
✓ Independent networking

Why important?
If one zone fails:
Other zones continue operating.

👉 **Availability Zone = Protection against datacenter failure**

🌟 **Benefits of Availability Zones**
✓ High availability
✓ Fault isolation
✓ Better reliability
✓ Disaster recovery within region

A region with availability zones usually has:
👉 **Minimum 3 Availability Zones**

Azure Service Categories for Availability Zones:

1️⃣ **Zonal Services:** Resource deployed to a specific availability zone.

Examples
- Virtual Machines
- Managed Disks
- Public IP Addresses

👉 **Zonal = Fixed to one zone**

2️⃣ **Zone-Redundant Services:**
Data/services automatically replicated across zones.

Examples
- Zone-redundant Storage
- SQL Database

👉 **Zone-Redundant = Replicated across zones**

3️⃣ **Non-Regional Services:** Services not tied to a specific region.

Examples
- Azure DNS
- Azure Traffic Manager
- Microsoft Entra ID

👉 **Non-Regional = Available globally**

🔁 **Region Pairs (VERY IMPORTANT)**
Two Azure regions paired together within the same geography.

Example
- East US ↔ West US

Purpose
Provides disaster recovery and replication.

Benefits
✓ Disaster recovery
✓ Replication support
✓ Prioritized restoration
✓ Reduced downtime

👉 **Region Pair = Backup region for another region**

If an entire Azure region fails:
👉 Region Pair helps provide recovery capability

🏛️ **Sovereign Regions:** Special Azure regions isolated for legal, compliance, or government requirements.

Examples
- US Government Regions
- China Regions

Purpose
Meet strict regulatory requirements.

👉 **Sovereign Region = Special compliance-focused Azure region**

🌟 **MOST IMPORTANT EXAM POINTS**
Azure hierarchy
👉 Geography → Region → Availability Zone → Datacenter

What is a Region?
👉 One or more datacenters in a geographic area

What is an Availability Zone?
👉 Physically separate datacenters within a region

Why use Availability Zones?
👉 High availability and fault tolerance

What is a Region Pair?
👉 Two regions linked for disaster recovery

What is a Datacenter?
👉 Physical facility containing servers and infrastructure

💔 **Example:**
Suppose a company deploys an application in:
Central India Region
Azure may place resources across:

- Availability Zone 1
- Availability Zone 2
- Availability Zone 3

If one datacenter/zone fails:
The application continues running from the other zones.
For even larger disasters, Azure can use a paired region for recovery.

Azure Management Infrastructure:

Azure Management Infrastructure: Azure Management Infrastructure is the hierarchy used to organize, manage, secure, and control Azure resources.

👉 **Azure uses a hierarchical structure to organize resources, billing, and access control.**

⏳ **Azure Management Hierarchy**

```
Management Group
      ↓
Subscription
      ↓
Resource Group
      ↓
Resource
```

👉 **Management Group → Subscription → Resource Group → Resource**

📦 **Azure Resource:** A resource is any service you create in Azure.

Examples

- Virtual Machine (VM)
- Storage Account
- Virtual Network (VNet)
- SQL Database

👉 **Resource = Actual Azure service**

📁 **Resource Group (VERY IMPORTANT)**
A logical container that holds Azure resources.

Why Important?
Helps organize related resources together.

Example
A web application may have:

- VM
- Database
- Storage Account
- VNet

inside one Resource Group.

⭐ **Resource Group Rules**
Rule 1
Every resource belongs to:
👉 Exactly ONE Resource Group

Rule 2
A Resource Group can contain:
👉 Multiple Resources

Rule 3
Resource Groups cannot be nested.

Rule 4
Deleting a Resource Group deletes all resources inside it.

Rule 5
Resource Groups cannot be renamed after creation.

If you delete a Resource Group:
👉 **All contained resources are deleted**.

💳 **Azure Subscription:** A subscription is a unit for:

- Billing
- Access Control
- Resource Management

Why Important?
Resources are created inside subscriptions.

Subscription Boundaries:

1️⃣ Billing Boundary

Controls:

- Billing
- Invoices
- Cost tracking

2️⃣ Access Control Boundary

Controls:

- Permissions
- Access policies
- User roles

👉 Subscription = Billing + Access Management

Why Create Multiple Subscriptions?

Environment Separation
Examples:

- Development
- Testing
- Production

Cost Tracking:
Separate billing for projects or teams.

Security:
Different access policies for different teams.

🏢 Management Groups (VERY IMPORTANT):

A container above subscriptions.
Used to manage multiple subscriptions together.

Why Important?

Apply policies and permissions across many subscriptions at once.

Example:-
Company Structure:
Marketing -

- Web Subscription
- Mobile Subscription

IT-

- Apps Subscription
- Infrastructure Subscription

👉 Management Group = Container for subscriptions

⭐ Management Group Benefits

✓ Centralized governance
✓ Centralized policy management
✓ Centralized access control
✓ Better organization

Important Facts:

Maximum nesting -
👉 6 levels deep
(Excluding root and subscription levels)

Parent Rule:
Each:

- Subscription
- Management Group

can have only ONE parent.

🌟 MOST IMPORTANT EXAM POINTS

What is a Resource?
👉 Any Azure service you create

What is a Resource Group?
👉 Logical container for resources

How many Resource Groups can a resource belong to?
👉 One

What happens if Resource Group is deleted?
👉 All resources inside are deleted

What is a Subscription?
👉 Billing and access-control boundary

What is a Management Group?
👉 Container for multiple subscriptions

Example: Suppose a company has:

Management Group: IT
Subscription: Production
Resource Group: WebApp-RG
Contains:

- VM
- Storage Account

- SQL Database
- VNet

If the company wants a security policy applied to all IT subscriptions:
👉 Apply it at the **Management Group** level.
If they delete **WebApp-RG**:
👉 VM, Database, Storage, and VNet are deleted together.

Component	Purpose
Management Group	Manage multiple subscriptions
Subscription	Billing & Access Control
Resource Group	Organize resources
Resource	Actual Azure service

Azure Compute Services:

Azure Virtual Machines (VMs): Azure Virtual Machines (VMs) are virtualized servers running in Azure cloud.
They provide Infrastructure as a Service (IaaS), allowing you to create and manage Windows or Linux servers.

❓ Why is it Important?
VMs allow organizations to:
✓ Run applications
✓ Host websites
✓ Create test environments
✓ Extend on-premises datacenters
✓ Migrate existing servers to cloud

⭐ Core Idea
👉 Azure VM = A server running in Microsoft's cloud

🧮 What Can You Control in a VM?
Since VM is an IaaS service, you manage:
✓ Operating System
✓ Installed Software
✓ Security Configuration
✓ User Accounts
✓ Applications

Azure manages:
✓ Physical Servers
✓ Datacenter
✓ Storage Infrastructure
✓ Physical Networking

👉 VM = Full server control in the cloud

🌟 Common Use Cases:
1️⃣ Testing & Development
Create temporary servers for testing applications.

2️⃣ Application Hosting
Run websites and applications.

3️⃣ Lift-and-Shift Migration
Move existing on-premises servers to Azure.

4️⃣ Disaster Recovery
Maintain backup workloads in Azure.

5️⃣ Datacenter Extension
Extend local infrastructure into cloud.

👉 VM = Rent a cloud computer/server

⚙️ VM Configuration Components
When creating a VM, you typically choose:
Compute
- vCPU
- RAM

Storage
- OS Disk
- Data Disk

Networking
- Virtual Network (VNet)
- Public IP
- Security Rules

🌟 VM Size Families (Exam Important):
Azure groups VMs into families.
You do NOT need to memorize every size.
Just remember the purpose.

Family	Purpose
B-Series	Budget / Burstable
D-Series	General Purpose
E-Series	Memory Optimized
F-Series	Compute Optimized

M-Series	Large Memory
L-Series	Storage Optimized
N-Series	GPU Workloads

⚙️ **VM Size Naming**
ExampleStandard_D2s_v5
Breakdown:

Part	Meaning
D	Family
2	Number of vCPUs
s	Premium SSD Support
v5	Hardware Generation

🌟 **VM Scaling**
Azure VMs can scale in two ways:

1️⃣ **Vertical Scaling**
Increase:
- CPU
- RAM

on the same VM.

👉 **Bigger VM**

2️⃣ **Horizontal Scaling**
Add multiple VM instances.

👉 **More VMs**

🌟 **Virtual Machine Scale Sets (VERY IMPORTANT):** Group of identical VMs that automatically scale.

Benefits
✓ Auto-scaling
✓ High Availability
✓ Load Balancing
✓ Automatic VM management

👉 **Scale Set = Auto-scaling VM group**

🌟 **Availability Sets:** Feature that improves VM availability inside a region.

Purpose
Protect against:
✓ Hardware failure
✓ Maintenance events

Uses
- Fault Domains
- Update Domains

👉 **Availability Set = VM redundancy inside a region**

🌟 **MOST IMPORTANT EXAM POINTS**
Azure VM belongs to which cloud model?
👉 IaaS

Who manages OS in Azure VM?
👉 Customer

Which VM family is memory optimized?
👉 E-Series

Which VM family is GPU optimized?
👉 N-Series
What is a VM Scale Set?
👉 Auto-scaling group of identical VMs

What improves VM resiliency?
👉 Availability Sets

❤️‍🩹**Example:** Suppose a company hosts an e-commerce website.
They create:
- 2 Linux VMs
- 1 Database
- 1 Storage Account

During a festival sale:
Traffic increases.
Azure Scale Set automatically creates additional VM instances.
When traffic decreases:
Extra VMs are removed.
This is cloud scalability in action.

Azure Virtual Desktop:

Azure Virtual Desktop (AVD): Azure Virtual Desktop (AVD) is a cloud-based desktop virtualization service that allows users to

securely access Windows desktops and applications from anywhere.

❓ **Why is it Important?**
Organizations can provide employees with:
✓ Remote desktops
✓ Remote applications
✓ Secure work environments
✓ Access from multiple devices
without needing a physical office PC.

👉 **Azure Virtual Desktop = Your Windows desktop running in Azure instead of your local computer**.

🌟 **Key Features**
✓ Access desktops from anywhere
✓ Centralized management
✓ Secure remote access
✓ Multi-device support
✓ Integration with Microsoft Entra ID

👉 **Desktop in the cloud**

📱 Supported Devices
Users can connect from:

- Laptop
- Desktop
- Tablet
- Mobile Phone
- Web Browser

🌟 **Main Components**
Users:
People accessing virtual desktops.

Microsoft Entra ID
Provides:

- Authentication
- Identity Management
- Access Control

Host Pool:
Collection of virtual machines that provide desktop sessions.

Session Host:
The VM that actually runs the desktop/application.

🧠 Simple Flow

```
User
 ↓
Microsoft Entra ID
 ↓
Azure Virtual Desktop
 ↓
Windows Desktop/App
```

🌟 **Common Use Cases**
1️⃣ Remote Work
Employees work securely from home.

2️⃣ Contractors & Temporary Staff
Provide temporary desktop access without issuing company laptops.

3️⃣ Call Centers / Support Teams
Standardized desktop environment for all employees.

4️⃣ Educational Institutions
Students access labs/software remotely.

🌟 **Benefits**
✓ Centralized desktop management
✓ Improved security
✓ Reduced data exposure
✓ Easy deployment
✓ Consistent user experience

🔒 **Security Benefit**
Data remains in Azure instead of being stored on employee devices.

👉 **Data stays in cloud, not on local device**

🌟 **MOST IMPORTANT EXAM POINTS**
What is Azure Virtual Desktop?
👉 Desktop and application virtualization service in Azure

Main purpose of AVD?
👉 Provide secure remote desktop access

What service handles identity and authentication?
👉 Microsoft Entra ID

Where do desktops run?
👉 Azure cloud

What is a Host Pool?
👉 Collection of session hosts (VMs)

💔 **Example:** Suppose a company has 500 employees working from home.
Instead of:
❌ Managing 500 individual office PCs
They create:
✓ Azure Virtual Desktop environment
Employees log in using:

- Laptop
- Tablet
- Browser

and access the same company desktop from anywhere.
The applications and data remain inside Azure, improving security and management.

Azure Virtual Desktop = Cloud-hosted Windows desktop and applications accessible securely from anywhere

Azure Containers: Containers are lightweight environments that package an application and its dependencies together so it can run consistently anywhere.

❓ **Why are Containers Important?**
Traditional Virtual Machines require:
❌ Full Operating System
❌ More resources
❌ Longer startup time
Containers provide:
✓ Faster startup
✓ Less resource usage
✓ Easy scaling
✓ Portability

⭐ **Core Idea**

👉 Containers package applications without requiring a full separate OS for each application.

👉 **Containers = Lightweight application packages**

⚖️ **Virtual Machines vs Containers**

Virtual Machine	Container
Full OS per VM	Shared host OS
Heavyweight	Lightweight
Slower startup	Faster startup
More resources	Fewer resources
Infrastructure focused	Application focused

🐬 **Docker:** Docker is the most popular container platform.

Purpose
Creates, packages, and runs containers.

👉 **Docker = Technology used to create containers**

🌟 **Azure Container Services**

1️⃣ Azure Container Instances **(ACI)**
Fastest and simplest way to run containers in Azure.

Characteristics
✓ No VM management
✓ No cluster management
✓ Quick deployment
✓ PaaS service

Best For

- Small workloads
- Testing
- Simple applications

👉 **ACI = Run a container quickly**

2️⃣ **Azure Container Apps:** Managed platform for running containerized applications.

Features
✓ Built-in scaling
✓ Built-in load balancing
✓ No infrastructure management
✓ PaaS offering

Best For
- Modern cloud applications
- APIs
- Microservices

👉 **Container Apps = Managed containers with scaling**

3️⃣ **Azure Kubernetes Service (AKS)**
Managed Kubernetes service used to orchestrate and manage large numbers of containers.

Purpose
Handles:
- Deployment
- Scaling
- Updates
- Lifecycle management

of containers.

Best For
- Large applications
- Enterprise workloads
- Microservices architecture

👉 **AKS = Kubernetes management platform**

Service	Purpose
ACI	Run a single/simple container
Container Apps	Managed scalable containers
AKS	Large-scale container orchestration

🏗️ **Microservices:** Application architecture where an application is divided into smaller independent services.

Example -
E-commerce Application:
- Frontend Service
- Backend Service
- Database Service

Each runs separately.

Benefits:
✓ Easier updates
✓ Independent scaling
✓ Better maintainability

👉 **Microservices = Small independent application components**

🌟 **MOST IMPORTANT EXAM POINTS**
Why use containers?
👉 Lightweight, portable, and scalable application deployment

Which service is simplest for running containers?
👉 Azure Container Instances (ACI)

Which service provides managed container apps with scaling?
👉 Azure Container Apps

Which Azure service uses Kubernetes?
👉 Azure Kubernetes Service (AKS)

Which is lighter?
👉 Containers are lighter than VMs

❤️‍🩹 **Example:**
Imagine a shopping website:
Frontend:
Shows products
Backend:
Processes orders
Database:
Stores data
Instead of putting everything on one VM, each component runs in its own container.
Benefits:
✓ Scale only the backend when traffic increases
✓ Update frontend independently
✓ Better performance and flexibility

This is why containers became so popular in cloud computing.

Containers = Lightweight application packaging

ACI = Simple containers

Container Apps = Managed containers

AKS = Kubernetes orchestration

Azure Functions: Azure Functions is a serverless compute service that allows you to run code in response to events without managing servers, VMs, or containers.

❓ **Why is it Important?**
With **traditional VMs:**
❌ Server must stay running
❌ You pay even when idle
With **Azure Functions:**
✓ Code runs only when needed
✓ Auto-scales automatically
✓ No server management
✓ Pay only when code executes

👉**Run code on demand without managing infrastructure**

👉**Azure Functions = Event-driven serverless compute**

⚡ **What is Serverless?**
Serverless does NOT mean there are no servers.
It means:
✓ Azure manages the servers
✓ You focus only on writing code

👉Serverless = No server management for developers

🔔 How Azure Functions Work:
Event Occurs
↓
Function Triggered
↓
Code Executes
↓
Function Stops
↓
Billing Stops

🌟 **Common Triggers:**
A function starts when an event occurs.

HTTP Trigger
Example: User visits a website or API.

Timer Trigger
Example: Run every day at 9 AM.

Queue Message Trigger
Example: Process incoming messages automatically.

Trigger = Event that starts a function.

🌟 **Common Outputs**
Functions can:
✓ Return API responses
✓ Write data to storage
✓ Send messages/events
✓ Update databases

🌟 **Benefits of Azure Functions**
✓ Serverless
✓ Auto-scaling
✓ Event-driven
✓ Fast deployment
✓ Cost-efficient
✓ No VM management

💰 **Cost Benefit**
You pay only while the function is running.
No cost for idle time.
👉 **Pay for execution time, not uptime**

🌟 **Automatic Scaling**
When demand increases:
Azure automatically creates more function instances.
When demand decreases:
Azure removes extra instances.

👉**Azure Functions automatically scale based on demand**

🌟 **Stateless vs Stateful**

Stateless (Default)
Every execution is independent.
No memory of previous execution.

Stateful (Durable Functions)
Function remembers previous state/workflow.
Used for long-running processes.

🌟 **Common Use Cases**

API Processing
Respond to web requests.

File Processing
Process uploaded files automatically.

Scheduled Tasks
Daily reports or backups.

Message Processing
React to queue messages.

Automation
Trigger actions automatically.

🌟 **MOST IMPORTANT EXAM POINTS**

Azure Functions belong to which compute model?
👉 Serverless Computing

What starts an Azure Function?
👉 Trigger/Event

Do you manage servers?
👉 No

When do you pay?
👉 Only when function runs

Does Azure Functions scale automatically?
👉 Yes

⚖️ Quick Comparison

Service	Use Case
VM	Full server control
Container	Run applications
Azure Functions	Run event-driven code
Azure Virtual Desktop	Cloud-hosted desktop

💔 **Example**

Imagine an online shopping website.
When a customer uploads a product image:

1. Image uploaded
2. Upload event occurs
3. Azure Function starts automatically
4. Image is resized
5. Function stops

Result:
✓ No server running 24/7
✓ Automatic scaling
✓ Pay only for actual processing
This is the power of serverless computing.

VM = Server

Container = Application Package

Azure Function = Event-Driven Serverless Code

Azure Virtual Desktop = Cloud Desktop

Azure AI, Machine Learning & IoT Services : Azure provides services for:
✓ Artificial Intelligence (AI)
✓ Machine Learning (ML)
✓ Internet of Things (IoT)
to help organizations build intelligent and connected solutions.

👉 **Use the right Azure service based on your business scenario**.

🤖 **Azure AI Services:** Prebuilt AI services that add intelligent features to applications without building or training your own AI model.

Why Important?
Developers can quickly add AI capabilities using APIs.

Common Capabilities
✓ Vision
✓ Speech
✓ Language Processing

✓ Document Intelligence

Examples
- Face recognition
- Speech-to-text
- Chatbots
- Text translation

👉 **Azure AI Services = Prebuilt AI APIs**

🤖 **Azure OpenAI Service:** Azure service that provides access to Generative AI models.

Examples
- Chat applications
- Content generation
- AI assistants
- Summarization

👉 **Azure OpenAI = Generative AI on Azure**

🧠**Agentic AI (AZ-900 Basic Understanding)**
Applications that combine:
- AI Model
- Instructions
- Context
- Tools

to complete tasks automatically.

Simple Example
An AI assistant that:
- Reads email
- Searches documents
- Creates a response

👉**Agentic AI = AI that performs multi-step tasks**

🧠 **Azure Machine Learning (Azure ML)**
A platform for building, training, deploying, and managing custom machine learning models.

Why Important?
Used when prebuilt AI services are not enough.

Example
A bank creates its own fraud detection model.

Common Tasks
✓ Train models
✓ Test models
✓ Deploy models
✓ Monitor models

👉 **Azure ML = Build your own AI model**

⚖️ **Azure AI vs Azure ML**

Azure AI Services	Azure Machine Learning
Prebuilt AI	Custom AI
Ready to use	Requires training
Faster setup	More flexibility
APIs	Model development

🌐 **Azure IoT Services:**
Internet of Things (IoT) refers to connected devices that send and receive data.

Examples:
- Smart sensors
- Smart meters
- Security cameras
- Industrial machines

👉 **IoT = Connected devices**

🌟**Azure IoT Hub:** Central communication service between devices and Azure.

Purpose:
Secure two-way communication.

👉**IoT Hub = Device communication center**

🌟**Azure IoT Central:** Managed SaaS platform for building IoT solutions quickly.

Purpose
Simplifies IoT deployment and management.

👉**IoT Central = Easy IoT management**

🌟 **Azure IoT Edge:** Allows processing and AI workloads to run directly on devices instead of the cloud.

Why Important?

Reduces latency and bandwidth usage.

Example
A factory machine analyzes data locally without constantly sending it to Azure.

👉IoT Edge = Cloud intelligence running near devices

🌟 Typical IoT Flow

```
Device
  ↓
IoT                                  Hub
  ↓
Cloud       Analytics       /       AI
  ↓
Insights & Actions
```

🌟 **MOST IMPORTANT EXAM POINTS**
Which service provides prebuilt AI APIs?
👉 Azure AI Services

Which service is used to build custom ML models?
👉 Azure Machine Learning

Which Azure service provides Generative AI?
👉 Azure OpenAI Service

Which service manages communication with IoT devices?
👉 Azure IoT Hub

Which service simplifies IoT solutions?
👉 Azure IoT Central

Which service brings cloud processing closer to devices?
👉 Azure IoT Edge

🧠 Easy Decision Framework
Need prebuilt AI?
➞ Azure AI Services

Need custom model training?
➞ Azure Machine Learning

Need connected devices?
➞ Azure IoT Services

💔 **Example**
Imagine a smart factory:
Machines
Send temperature data
↓
Azure IoT Hub
Collects device data
↓
Azure AI Services
Analyzes data
↓
Azure ML
Predicts equipment failure
↓
IoT Edge
Runs predictions directly on factory devices
This is how Azure combines AI, ML, and IoT into one intelligent solution.

Need	Service
Prebuilt AI	Azure AI Services
Generative AI	Azure OpenAI
Custom AI Model	Azure Machine Learning
Device Communication	IoT Hub
Easy IoT Platform	IoT Central
Processing on Device	IoT Edge

Application Hosting Options:
Azure provides multiple ways to host applications depending on how much control and management you want.

⭐ **Core Idea**
👉 More control = More management
👉 Less management = Less control

Service	Control	Management Effort
Virtual Machine	High	High
Containers	Medium	Medium
Azure App Service	Lower	Low

🖥️ 1️⃣ **Azure Virtual Machines:**

A cloud server where you manage the operating system and application.

You Manage
✓ OS
✓ Applications
✓ Updates
✓ Configuration

Best For
- Legacy applications
- Custom environments
- Full control requirements

👉 **VM = Maximum Control**

🐬 2️⃣ **Container:** Lightweight application packaging that runs consistently across environments.

Benefits
✓ Portable
✓ Fast deployment
✓ Easy scaling
✓ Microservices architecture

Best For
- Modern applications
- Scalable workloads
- DevOps environments

👉 **Containers = Application-focused hosting**

🌐 3️⃣ **Azure App Service (VERY IMPORTANT):** A fully managed platform for hosting web applications, APIs, and mobile backends.

Why Important?
You focus on the application.
Azure manages the infrastructure.

Benefits
✓ No server management
✓ Automatic scaling
✓ High availability
✓ Built-in security
✓ Continuous deployment

👉 **App Service = Host apps without managing servers**

VM
You manage everything.

Container
You manage the application package.

App Service
Azure manages most infrastructure.

🌟 **What Can Azure App Service Host?**

1️⃣ **Web Apps**
Purpose
Host websites and web applications.

Examples
- Company websites
- E-commerce sites
- Portals

👉 **Web App = Website hosting**

2️⃣ **API Apps**
Purpose
Host REST APIs.

Example
Backend service for a mobile app.

👉 **API App = Backend services**

3️⃣ **WebJobs**
Purpose
Run background tasks and scripts.

Examples
- Scheduled reports
- Automated processing
- Background jobs

👉 **WebJobs = Background tasks**

🅰️ Mobile Apps

Purpose
Provide backend services for mobile applications.

Features
✓ Authentication
✓ Push Notifications
✓ Cloud Database Access

👉 Mobile Apps = Mobile backend platform

🌟 Azure App Service Features

✓ Automatic Scaling
✓ Load Balancing
✓ High Availability
✓ CI/CD Integration
✓ Windows & Linux Support
✓ GitHub Integration
✓ Azure DevOps Integration

🧠 Exam:

Azure App Service supports:
✓ .NET
✓ Java
✓ Python
✓ Node.js
✓ PHP
and many other frameworks.

🌟 MOST IMPORTANT EXAM POINTS

Which service gives maximum control?
👉 Azure Virtual Machines

Which service is lightweight and portable?
👉 Containers

Which service is fully managed for hosting web apps?
👉 Azure App Service
Which App Service type hosts REST APIs?
👉 API Apps

Which App Service feature runs background tasks?
👉 WebJobs

⚖️ Quick Comparison

Feature	VM	Container	App Service
Control	High	Medium	Low
OS Management	You	Shared	Azure
Scaling	Manual	Easier	Automatic
Best For	Full control	Modern Apps	Web Apps & APIs

❤️‍🩹 Example:

Imagine you build a college placement portal.
Option 1: VM
You install:

- Windows/Linux
- Web Server
- Database tools

and manage everything yourself.

Option 2: Containers
You package the application into Docker containers and deploy it.

Option 3: Azure App Service
You simply upload the application code.
Azure handles:
✓ Servers
✓ Scaling
✓ Load balancing
✓ Availability
This is why many modern organizations prefer App Service for web applications.

Azure App Service = Fully Managed Web Hosting

Azure Networking Services:

Azure Virtual Network (VNet) is a private network in Azure that allows Azure resources to communicate securely.

❓ Why is it Important?

VNet enables:
✓ Resource communication
✓ Internet connectivity
✓ Secure cloud networking
✓ Hybrid cloud connectivity

✓ Traffic control

⭐ **Core Idea**
👉 VNet = Your private network inside Azure

👉 **VNet is the cloud version of a traditional network**

🌟 **Key Capabilities of VNet**
Azure VNet allows:
✓ Isolation & Segmentation
✓ Internet Communication
✓ Azure Resource Communication
✓ On-Premises Connectivity
✓ Traffic Routing
✓ Traffic Filtering
✓ VNet Peering

🏚️ **Isolation & Segmentation:** A VNet has its own private IP address space.
You can divide it into smaller networks called:
👉 Subnets

Example

```
VNet (10.0.0.0/16)

├── Subnet-A (10.0.1.0/24)
├── Subnet-B (10.0.2.0/24)
└── Subnet-C (10.0.3.0/24)
```

👉 **VNet = Network**
👉 **Subnet = Smaller section of network**

🌐 **Public vs Private Endpoints**
Public Endpoint
Has a Public IP.
Accessible from Internet.

Example
Website accessible globally.

Private Endpoint
Has a Private IP.
Accessible only within VNet or connected networks.

Example
Internal database server.

👉 **Public = Internet**
👉 **Private = Internal Network**

🌐 Internet Communication
Resources can communicate with the internet by:
✓ Public IP Address
✓ Load Balancer

Example
Public website hosted in Azure.

🔁 **Communication Between Azure Resources**
Azure resources inside a VNet can communicate securely.

Examples
- VM ↔ Database
- VM ↔ Storage Account
- App Service ↔ Database

👉 **Same VNet = Easy communication**

🏢 **Connect to On-Premises Network**
Azure provides three options.

1️⃣ Point-to-Site VPN
Single computer connects to Azure VNet.

Example
Employee working remotely.

👉 **One Device → Azure**

2️⃣ Site-to-Site VPN
Entire office network connects to Azure.

Example
Company datacenter connected to Azure.

👉 **Office ↔ Azure**

3️⃣ **ExpressRoute** (VERY IMPORTANT)
Private dedicated connection to Azure.
Does NOT use public internet.

Benefits

✓ More secure
✓ Higher bandwidth
✓ Lower latency

👉 **ExpressRoute = Private highway to Azure**

🌟 **MOST ASKED EXAM QUESTION**
Which connectivity option does NOT use the internet?
👉 ExpressRoute

🚦 Route Network Traffic
Azure automatically routes traffic.
You can customize routing using:
✓ Route Tables
✓ BGP
✓ User Defined Routes (UDR)

👉 **Routing = Decides packet path**

🔒 **Filter Network Traffic**
Azure allows traffic filtering using:

1️⃣ Network Security Groups (NSG) ⭐
Firewall-like rules controlling traffic.

Controls
✓ Allow traffic
✓ Deny traffic

Based On
- IP Address
- Port Number
- Protocol

Example
Allow:
- HTTPS (443)
- SSH (22)

Block:
- Everything else

👉 **NSG = Azure firewall rules**

2️⃣ Network Virtual Appliance (NVA)
Specialized VM performing networking tasks.

Examples
- Firewall
- IDS/IPS
- WAN Optimization

👉 **NVA = Advanced network appliance**

🔗 **VNet Peering** (VERY IMPORTANT)
Connects two Azure VNets directly.

Benefits
✓ Private communication
✓ High performance
✓ Microsoft backbone network
✓ No public internet

Example

```
East US VNet
      ↓
VNet Peering
      ↓
West Europe VNet
```

👉 **VNet Peering = Connect two VNets privately**

🌟 **MOST IMPORTANT EXAM POINTS**
What is Azure VNet?
👉 Private network in Azure

What divides a VNet?
👉 Subnets

Which connectivity option uses private dedicated connectivity?
👉 ExpressRoute

What filters traffic?
👉 Network Security Group (NSG)

What connects two VNets?
👉 VNet Peering

What is a subnet?
👉 Smaller network inside a VNet

❤️ **Example:**
Imagine a company deploys:

Azure VNet
10.0.0.0/16
Inside it:
Subnet-Web
Web Servers
Subnet-DB
Databases
NSG rules:
✓ Allow HTTPS
✓ Allow SSH
❌ Block unwanted traffic
The company office connects to Azure through:
👉 Site-to-Site VPN
Another Azure region connects using:
👉 VNet Peering
This is a typical real-world Azure network design.

The topics you should remember especially well are:

1. **VNet**
2. **Subnet**
3. **NSG**
4. **VPN**
5. **ExpressRoute**
6. **VNet Peering**
7. **Site-to-Site VPN**

Azure virtual private networks:

Azure Virtual Private Networks (VPN):
A VPN (Virtual Private Network) creates an encrypted tunnel between networks or devices over the internet.
It allows secure communication between Azure and external networks.

❓ Why is it Important?

VPN provides:
✓ Secure communication
✓ Data encryption
✓ Hybrid cloud connectivity
✓ Remote access

⭐ Core Idea

👉 VPN = Secure encrypted connection over the public internet

👉 **VPN protects data while it travels across the internet**

🌐 Azure VPN Gateway

Azure VPN Gateway is a service that connects Azure Virtual Networks to:

- Users
- Offices
- Datacenters
- Other Azure VNets

using encrypted VPN tunnels.

👉 **VPN Gateway = Entry/Exit point for VPN connections**

🌟 VPN Connectivity Types

1️⃣ Point-to-Site (P2S)
Connects a single device to Azure.

Example
Employee working from home.
Laptop → Azure VNet

👉 **One Device → Azure**

🌟 Use Cases

✓ Remote workers
✓ Administrators
✓ Individual users

2️⃣ Site-to-Site (S2S)
Connects an entire office/datacenter network to Azure.

Example
Company Office ↔ Azure

👉 **Office Network → Azure**

🌟 Use Cases

✓ Hybrid Cloud
✓ Datacenter Extension
✓ Enterprise Connectivity

3️⃣ Network-to-Network (VNet-to-VNet)
Connects one Azure Virtual Network to another Azure Virtual Network.

Example
Azure VNet A ↔ Azure VNet B

👉 **Azure Network ↔ Azure Network**

🌟 **VPN Gateway Types**

Policy-Based VPN:
How it works
Uses predefined policies to determine encrypted traffic.

Route-Based VPN ⭐:
How it works
Uses routing tables to determine traffic paths.

Benefits
✓ More flexible
✓ Preferred by Microsoft
✓ Supports more scenarios

👉 **Route-Based VPN = Recommended VPN type**

🌟 **High Availability Options**

1️⃣ Active / Standby
Default configuration.
One gateway is active.
Second gateway waits as backup.

If failure occurs
Backup automatically takes over.

👉 **One working + one backup**

2️⃣ Active / Active
Both gateway instances process traffic simultaneously.

Benefits
✓ Better performance
✓ Higher availability
✓ No single point of failure

👉 **Both gateways working**

3️⃣ Zone-Redundant Gateway
Gateway instances deployed across Availability Zones.

Benefit
Survives entire zone failure.

👉 **Protected against Availability Zone failure**

🌟 **VPN vs ExpressRoute (VERY IMPORTANT)**

VPN	ExpressRoute
Uses Internet	Private Connection
Lower Cost	Higher Cost
Encrypted Tunnel	Dedicated Circuit
Good for Most Companies	Enterprise Grade

🧠 **MOST ASKED EXAM QUESTION**
Which service does NOT use the public internet?
👉 ExpressRoute

🌟 MOST IMPORTANT EXAM POINTS
What is VPN?
👉 Secure encrypted tunnel over internet

What service provides VPN connectivity in Azure?
👉 Azure VPN Gateway

Point-to-Site connects?
👉 Single Device → Azure

Site-to-Site connects?
👉 Office Network → Azure

Network-to-Network connects?
👉 Azure VNet → Azure VNet

Which VPN type is preferred?
👉 Route-Based VPN

Which connectivity option does not use internet?

👉 ExpressRoute

💔**Example:**
Imagine a company:
Employees at Home
Use:
👉 Point-to-Site VPN
Company Office
Uses:
👉 Site-to-Site VPN

Azure East US Network
Connects to:
Azure West Europe Network
Using:
👉 Network-to-Network VPN
All communication is encrypted while travelling over the internet.

Azure ExpressRoute: Azure ExpressRoute provides a **private dedicated connection** between your on-premises network and Azure.
Unlike VPN, traffic does **not travel through the public internet**.

Why is it Important?
Provides:
✓ Higher Security
✓ Lower Latency
✓ Faster Speeds
✓ More Reliable Connectivity
✓ Predictable Performance

Core Idea
👉 ExpressRoute = Private Connection to Azure

👉 No Public Internet Involved

How ExpressRoute Works

```
Office / Datacenter
        ↓
Connectivity Provider
        ↓
ExpressRoute Circuit
        ↓
Microsoft Cloud
```

Traffic travels through Microsoft's private network.

ExpressRoute Circuit: A dedicated connection between:
Your Network ↔ Microsoft Network
Every ExpressRoute connection is called an:
👉 ExpressRoute Circuit

👉 **Circuit = Private path to Azure**

Features & Benefits
1. Private Connectivity
Traffic never travels through the public internet.

2. High Reliability
Built-in redundancy.
Multiple backup paths available.

3. Low Latency
Faster response times.

4. High Throughput
Suitable for large data transfers.

5. Better Security
Reduced exposure to internet threats.

Global Reach
What is it?
ExpressRoute Global Reach allows different company locations to communicate using Microsoft's backbone network.

Example

```
India Office
      ↓
Microsoft Network
      ↓
Europe Office
```

Without using the public internet.

👉 **Global Reach = Connect offices through Microsoft network**

Dynamic Routing
ExpressRoute uses:
👉 BGP (Border Gateway Protocol)

What does BGP do?
Automatically exchanges routing information between:

- Azure
- On-premises networks

👉 **ExpressRoute uses BGP**

Built-in Redundancy
Microsoft provides redundant connectivity.
If one path fails:
✓ Backup path takes over
✓ High availability maintained

When Should You Use ExpressRoute?
Use ExpressRoute when:
✓ High security required
✓ Compliance requirements exist
✓ Large data transfer needed
✓ Low latency required
✓ Mission-critical workloads
✓ Avoiding public internet is important

Security Consideration
Key Point:
Traffic does not pass through the public internet.
This reduces:

- Exposure to attacks
- Internet congestion
- Latency fluctuations

VPN vs ExpressRoute ⭐ ⭐ ⭐

Feature	VPN	ExpressRoute
Internet Used	Yes	No
Private Connection	No	Yes
Cost	Lower	Higher
Speed	Good	Very High
Security	High	Very High
Reliability	Good	Excellent
Latency	Variable	Predictable

Exam Questions
Q1. Which service provides private connectivity to Azure?
✅ ExpressRoute

Q2. Which service does NOT use the public internet?
✅ ExpressRoute

Q3. Which service is cheaper?
✅ VPN

Q4. Which service uses BGP?
✅ ExpressRoute

Q5. Which service is best for compliance-sensitive workloads?
✅ ExpressRoute

AZ-900 Exam Snapshot
Remember only these 5 points:

1. ExpressRoute = Private connection to Azure
2. Does NOT use public internet
3. Uses BGP routing
4. More secure and reliable than VPN
5. Expensive but enterprise-grade

Real-World Example
Imagine a bank.
The bank:

- Has sensitive customer data
- Needs low latency
- Must satisfy compliance rules

Using a normal VPN may not be enough.
So the bank uses:
👉 **Azure ExpressRoute**
because traffic remains on a dedicated private connection rather than the public internet.

Azure DNS: Azure DNS is a cloud service that hosts and manages DNS domains using Azure infrastructure.

It translates:
www.google.com
↓
`142.250.x.x`
Human-readable names → IP addresses

Why is it Important?
Humans remember:

www.microsoft.com
Computers communicate using:
`20.112.52.29`
DNS acts as the translator.

👉 **DNS = Name Resolution Service (Name → IP Address)**

Benefits of Azure DNS
1. Reliability & Performance
Uses Microsoft's global DNS infrastructure.
Automatically routes DNS requests to the nearest healthy DNS server.
Benefits:
✓ Fast response
✓ High availability
✓ Global reach

2. Security
Protected by Azure's secure infrastructure.
Integrated with Azure authentication and management.

3. Ease of Use
Manage DNS through:
- Azure Portal
- Azure CLI
- PowerShell
- REST API

4. Private DNS Support
Supports private DNS zones inside VNets.
Example:
`database.company.local`
Accessible only within your private network.

5. Alias Records
One of the favourite AZ-900 concepts.
Alias records allow DNS names to point directly to Azure resources.
Examples:
- Public IP
- Traffic Manager
- CDN Endpoint

Why Alias Records?
Normally:
`DNS Record → IP Address`
Problem:
If IP changes → DNS record must be updated manually.

With Alias Record:
`DNS Record → Azure Resource`
If resource IP changes:
✓ DNS updates automatically
✓ No manual changes needed

👉 **Alias Record = Automatic DNS update**

Important Exam Point
Azure DNS Cannot Buy Domains
Azure DNS only hosts and manages DNS records.
You must buy domain names separately from:
- GoDaddy
- Namecheap
- Other registrars

Then host DNS records in Azure DNS.

AZ-900 Exam Trap
❌ Azure DNS purchases domain names
✅ Azure DNS manages DNS records

Practical Example
Suppose you own:
`mycompany.com`
You purchase it from:
`Namecheap`
Then host DNS records in:
`Azure DNS`
When users visit:
www.mycompany.com
Azure DNS resolves the domain to the correct Azure resource.

Quick Exam Questions
Q1. What is Azure DNS used for?
✅ Name resolution

Q2. DNS converts?
✅ Domain names → IP addresses

Q3. Can Azure DNS purchase domains?
❌ No

Q4. What record automatically updates when Azure resource IP changes?

✅ Alias Record

Q5. What is a private DNS zone used for?
✅ Internal name resolution inside a VNet

DNS -> Name – IP resolution

Azure Storage Services:

Azure Storage Account: A Storage Account is a container that stores Azure storage services.
Think of it as:

```
Storage Account
      ↓
Contains
      ↓
Blobs, Files, Queues, Tables
```

Why is it Important?
A Storage Account provides:
✓ Secure Storage
✓ High Availability
✓ Durability
✓ Massive Scalability
✓ Global Accessibility (HTTP/HTTPS)

👉 **Storage Account = Parent Container for Azure Storage Services**

Azure Storage Services

Service	Purpose
Blob Storage	Store files and unstructured data
Azure Files	Shared file storage
Queue Storage	Message queues
Table Storage	NoSQL structured data

1. Blob Storage
Stores:

- Images
- Videos
- Documents
- Backups
- Logs

Example

```
Photos
Videos
PDFs
ZIP Files
```

stored in Blob Storage.

👉 **Blob = Unstructured Data Storage**

2. Azure Files: Managed file shares in Azure.
Supports:

- SMB
- NFS

Example
A company stores shared documents:

```
HR Files
Policies
Reports
```

accessible by multiple users.

👉 **Azure Files = Shared Network Drive in Cloud**

3. Queue Storage: Stores messages between applications.

Example

```
Website
   ↓
Queue
   ↓
Background Worker
```

Processes tasks later.

👉 **Queue = Messaging Service**

4. Table Storage: Stores structured NoSQL data.

Example

```
Customer Data
Sensor Data
User Profiles
```

without a traditional SQL database.

👉 **Table = NoSQL Data Storage**

Azure Storage Account Types
Standard GPv2 ⭐ (Most Important)
Recommended for most workloads.
Supports:
✓ Blob

✓ Files
✓ Queues
✓ Tables

Exam Tip
👉 GPv2 = Default Choice

Premium Block Blobs
Used for:
- High transaction workloads
- Low latency storage

Premium File Shares
Used for:
- High-performance file shares
- SMB/NFS workloads

Premium Page Blobs
Used mainly for:
- VM disks

Quick Memory
`GPv2 → General Purpose`

`Premium Block → Fast Blob Storage`

`Premium File → Fast File Shares`

`Premium Page → VM Disks`

Storage Redundancy Options

Type	Meaning
LRS	Locally Redundant Storage
ZRS	Zone Redundant Storage
GRS	Geo Redundant Storage
GZRS	Geo-Zone Redundant Storage
RA-GRS	Read Access Geo Redundant
RA-GZRS	Read Access Geo-Zone Redundant

👉 **More letters = More Redundancy = More Availability**

Storage Account Naming Rules ⭐
A storage account name:
✓ Must be globally unique
✓ 3–24 characters
✓ Lowercase letters only
✓ Numbers allowed

Valid Example
`mystorage123`

Invalid Example
`MyStorage`
(Capital letters not allowed)

Storage Account Endpoints
Each storage service gets its own endpoint.

Storage service	Endpoint
Blob Storage	https://<storage-account-name>.blob.core.windows.net
Data Lake Storage Gen2	https://<storage-account-name>.dfs.core.windows.net
Azure Files	https://<storage-account-name>.file.core.windows.net
Queue Storage	https://<storage-account-name>.queue.core.windows.net
Table Storage	https://<storage-account-name>.table.core.windows.net

Just remember:
`Blob → .blob.`
`Files → .file.`
`Queue → .queue.`
`Table → .table.`

Quick Exam Questions
Q1. Which storage service stores images and videos?
✅ Blob Storage

Q2. Which service provides cloud file shares?
✅ Azure Files

Q3. Which service stores messages?
✅ Queue Storage

Q4. Which service stores NoSQL data?
✅ Table Storage

Q5. Recommended storage account type for most workloads?
✅ Standard GPv2

Q6. Can two storage accounts have the same name?
❌ No

Azure Storage Redundancy:

What is Redundancy?
Keeping multiple copies of data to prevent data loss.
Azure automatically creates copies of your data.
Why is it Important?
Protects against:
✓ Disk failures
✓ Server failures
✓ Datacenter failures
✓ Availability Zone failures
✓ Regional disasters

More Copies = More Protection = More Cost

Types of Redundancy:
Azure Storage offers:

1. LRS
2. ZRS
3. GRS
4. GZRS
5. RA-GRS
6. RA-GZRS

1. LRS (Locally Redundant Storage)
Azure keeps:
Copy 1
Copy 2
Copy 3
inside the SAME datacenter.

Protection Against
✅ Disk failure
✅ Server failure

Does NOT Protect Against
❌ Datacenter failure
❌ Regional disaster

Copies
3 Copies
1 Datacenter

Cost
💲 Cheapest

👉 **LRS = 3 copies in ONE datacenter**

2. ZRS (Zone Redundant Storage)
Azure keeps copies across:
Availability Zone 1
Availability Zone 2
Availability Zone 3
inside the same region.
Protection Against
✅ Server failure
✅ Datacenter failure
✅ Availability Zone failure

Does NOT Protect Against
❌ Regional disaster

Copies
3 Copies
3 Availability Zones
1 Region

👉 **ZRS = Survives Zone Failure**

LRS vs ZRS

Feature	LRS	ZRS
Copies	3	3
Datacenter	One	Multiple Zones
Zone Failure Protection	No	Yes
Cost	Lower	Higher

3. GRS (Geo-Redundant Storage)
Azure stores:
Primary Region
3 Copies (LRS)
Secondary Region
3 Copies (LRS)

Total Copies
6 Copies
2 Regions

Protection Against
✅ Regional disaster

✅ Datacenter failure
✅ Server failure

Replication Type
Primary → Secondary
`Asynchronous`

👉 **GRS = LRS + Second Region**

4.GZRS(Geo-Zone Redundant Storage)
Combines:
`ZRS + Geo Replication`

Primary Region
`3 Zones (ZRS)`
Secondary Region
`3 Copies (LRS)`

Protection Against
✅ Zone failure
✅ Datacenter failure
✅ Regional disaster

Highest Resiliency: Microsoft recommends GZRS for critical workloads.
👉 **GZRS = Best Protection**

Read Access Versions
Normally:
Secondary region is: `Backup Only`
You cannot read from it.

RA-GRS(Read Access Geo Redundant Storage)
Allows reading from:
`Primary Region`
`+`
`Secondary Region`

RA-GZRS
Allows reading from:
`Primary Region`
`+`
`Secondary Region`
while also having zone redundancy.

👉 **RA = Read Access to Secondary Region**

Easy Memory Formula

```
LRS = Local

ZRS = Zones

GRS = Geo

GZRS = Geo + Zones

RA = Read Access
```

Durability (Exam Only)

Type	Durability
LRS	11 nines
ZRS	12 nines
GRS	16 nines
GZRS	16 nines

Just remember:
`LRS < ZRS < GRS ≈ GZRS`

```
LRS
↓
OneDatacenter

ZRS
↓
Three Zones

GRS
↓
Two Regions

GZRS
↓
Three Zones + Two Regions
```

Exam Questions
Q1. Cheapest redundancy option?
✅ LRS

Q2. Protects against Availability Zone failure?
✅ ZRS

Q3. Protects against Regional disaster?
✅ GRS

Q4. Highest resiliency option?
✅ GZRS

Q5. Which option allows reading from secondary region?
✅ RA-GRS
✅ RA-GZRS

Q6. What does RA stand for?
✅ Read Access

Remember only these 5 lines:

```
LRS  = 3 copies, one datacenter

ZRS  = 3 copies, three zones

GRS  = 6 copies, two regions

GZRS = Zones + Regions

RA   = Read access to secondary region
```

For Cloud Career

As someone targeting **Azure Administration / Infrastructure / Cloud Support**, this is more than exam theory. When creating a Storage Account in Azure Portal, one of the first decisions you'll make is:

LRS
ZRS
GRS
GZRS

In real companies:

- Development/Test → LRS
- Business Applications → ZRS
- Critical Production Systems → GRS/GZRS

Azure Storage Services:

Why Azure Storage?

Azure Storage is:
✓ Durable
✓ Highly Available
✓ Secure
✓ Scalable
✓ Managed by Microsoft
✓ Accessible globally

👉**Azure Storage = Secure + Scalable + Highly Available**

1. Azure Blob Storage ⭐ ⭐ ⭐

What is it?
Blob Storage stores:

```
Images
Videos
PDFs
Backups
Logs
Documents
```

Unstructured data.

Why is it Used?

For:
✓ Website images
✓ Video streaming
✓ Backup storage
✓ Log files
✓ Data lakes

👉 **Blob = Unstructured Data**

Blob Access Tiers ⭐ ⭐ ⭐

Azure provides four access tiers.

Hot Tier

Used when data is accessed frequently.
Example

```
Website                          Images
Active Documents
```

Characteristics
✓ Highest storage cost
✓ Lowest access cost
✓ Fastest access

👉 **Hot = Frequently Used**

Cool Tier

Used when data is accessed occasionally.
Minimum storage:

```
30 Days
```

Example

```
Invoices
Reports
```

👉 **Cool = Sometimes Used**

Cold Tier
Used for rarely accessed data.
Minimum storage:
`90 Days`

👉 **Cold = Rarely Used**

Archive Tier ⭐⭐⭐
Used for long-term storage.
Minimum storage:
`180 Days`

Example
`Old Backups`
`Compliance Data`
`Historical Records`

Characteristics
✓ Cheapest storage
✓ Slowest retrieval

👉 **Archive = Long-Term Storage**

```
Hot
 ↓
Cool
 ↓
Cold
 ↓
Archive
```

As usage decreases:
✓ Storage cost decreases
❌ Retrieval time increases

2. Azure Files ⭐⭐⭐
What is it?
Managed cloud file shares.
Supports:

- SMB
- NFS

Real Example
Shared company folder:
`HR Documents`
`Projects`
`Reports`
accessible by many users.

Benefits
✓ Shared access
✓ Cloud + On-premises
✓ Fully managed
✓ Windows/Linux support

👉 **Azure Files = Shared Network Drive**

3. Azure Queue Storage
What is it?
Stores messages between applications.
Supports asynchronous processing.

Example
`Website`
`↓`
`Queue`
`↓`
`Azure Function`

Key Fact
Message size:
`64 KB`

👉 **Queue = Messaging Service**

4. Azure Disks ⭐⭐⭐
What is it?
Provides storage for Azure Virtual Machines.

Example
`Windows VM Disk`
`Linux VM Disk`

Characteristics
✓ Block-level storage
✓ Managed by Azure
✓ High performance

👉 **Azure Disks = VM Storage**

5. Azure Tables

What is it?
NoSQL storage for structured data.

Example
`Customer Data`
`IoT Data`
`Product Catalog`

Characteristics
✓ Non-relational
✓ Highly scalable
✓ Schema flexible

👉 **Tables = NoSQL Storage**

Exam Questions
Q1. Best storage for images and videos?
✅ Blob Storage

Q2. Best storage for shared folders?
✅ Azure Files

Q3. Best storage for VM operating system disks?
✅ Azure Disks

Q4. Best service for application messaging?
✅ Queue Storage

Q5. Best service for NoSQL structured data?
✅ Azure Tables

Q6. Cheapest Blob tier?
✅ Archive

Q7. Most frequently accessed tier?
✅ Hot

Blob Tier Comparison ⭐ ⭐ ⭐

Tier	Access Frequency	Minimum Storage
Hot	Frequent	None
Cool	Occasional	30 Days
Cold	Rare	90 Days
Archive	Very Rare	180 Days

`Hot → Cool → Cold → Archive`

Identify Azure data Migration Options:

Azure Data Migration Options:
Azure provides two major ways to move data and workloads to Azure:

Service	Purpose
Azure Migrate	Online migration
Azure Data Box	Offline migration

Why Data Migration Matters
When moving from:
`On-Premises → Azure Cloud`
you need a way to transfer:

- Servers
- Databases
- Applications
- Files
- Large datasets

Azure provides specialized migration tools for this purpose.

1. Azure Migrate ⭐ ⭐ ⭐
Azure Migrate is Microsoft's central migration hub.
It helps you:
✓ Discover resources
✓ Assess readiness
✓ Migrate workloads
✓ Track migration progress

Think of It As
`Migration Control Center`
for moving everything to Azure.

Main Functions
Discovery
Finds:

- Servers
- Applications
- Databases

inside your current environment.

Assessment
Checks:

- Compatibility
- Azure readiness
- Cost estimates
- Resource requirements

Migration
Moves workloads into Azure.
Examples:

- Virtual Machines
- Databases
- Web Apps

👉 **Azure Migrate = Assess + Migrate**

Integrated Migration Tools
Azure Migrate supports:
Server Migration
Move:

```
Physical                    Servers
VMware                        VMs
Hyper-V VMs
```

to Azure VMs.

Database Migration
Move:

```
SQL                         Server
MySQL
PostgreSQL
```

to Azure.

Web App Migration
Move websites and web applications.

👉**Azure Migrate = One Portal for Cloud Migration**

Azure Migrate Workflow

```
Discover
   ↓
Assess
   ↓
Migrate
   ↓
Monitor
```

Example
A company has:

```
50                          Servers
10                        Databases
20 Applications
```

running in its datacenter.
They want to move everything to Azure.
Use:
✅ Azure Migrate
because migration occurs online through the network.

2. Azure Data Box ⭐⭐⭐
Azure Data Box is a physical migration appliance.
Microsoft ships a secure storage device to you.
You copy data onto it and send it back.
Microsoft uploads the data into Azure.

Why Use It?
When internet transfer is:
❌ Too slow
❌ Too expensive
❌ Too large

Capacity
Azure Data Box supports up to:

```
80 TB
```

of usable storage.

👉 **Data Box = Ship Data Physically**

Data Box Workflow

```
Order             Data              Box
        ↓
Receive                          Device
        ↓
Copy                              Data
        ↓
Ship                              Back
        ↓
Microsoft Uploads to Azure
```

Common Use Cases
One-Time Migration
Move:

```
50                                  TB
100                                 TB
200 TB
```

of historical data.

Periodic Large Uploads
When data is generated faster than internet bandwidth can handle.

Export Data From Azure
Download large amounts of Azure data back to your datacenter.
👉 **Huge Data + Slow Network = Data Box**

Security Features
Azure Data Box provides:
✓ End-to-end tracking
✓ Secure transport
✓ Encrypted storage
✓ Disk wiping after use

NIST Compliance
After transfer:
Disks are securely erased using NIST 800-88 standards.

Azure Migrate vs Azure Data Box ⭐⭐⭐

Feature	Azure Migrate	Azure Data Box
Migration Type	Online	Offline
Transfer Method	Internet	Physical Device
Best For	Servers, Apps, Databases	Massive Data Transfers
Assessment Tools	Yes	No
Real-Time Migration	Yes	No
Large Dataset Support	Limited by bandwidth	Excellent

If question mentions:

Servers
Applications
Databases
Assessment
Migration Planning
✅ Azure Migrate

If question mentions:
50 TB+
Slow Internet
Physical Device
Bulk Transfer
✅ Azure Data Box

Exam Questions

Q1. Which service assesses on-premises resources before migration?
✅ Azure Migrate

Q2. Which service uses a physical appliance?
✅ Azure Data Box

Q3. Best option for migrating 60 TB of data with slow internet?
✅ Azure Data Box

Q4. Best tool to migrate VMs and databases to Azure?
✅ Azure Migrate

Q5. What is the maximum usable capacity of Azure Data Box?
✅ 80 TB

Azure Migrate
= Discover + Assess + Migrate
Azure Data Box
= Physical Device for Huge Data Transfers

Identify Azure File Movements Options:

Azure provides tools for moving and synchronizing files between:

- Local systems
- Azure Storage
- Different storage accounts
- Hybrid environments

The three main file movement tools are:

1. AzCopy ⭐⭐⭐

AzCopy is a command-line utility used to transfer data to and from Azure Storage.
It can:
✓ Upload files
✓ Download files
✓Copy data between storage accounts
✓ Synchronize files
✓ Move data between cloud providers

Think of It As
Robocopy for Azure Storage
but cloud-enabled.

Common Uses

- Bulk uploads
- Bulk downloads
- Backup transfers

- Automated scripts
- Scheduled file movement

Synchronization Behavior
Important Exam Point
AzCopy synchronization is:
One-Way Sync
Source → Destination
NOT:
Bi-directional Sync

Example
Local Folder
↓
Azure Blob Storage
Changes flow only in the chosen direction.

Why Use AzCopy?
Advantages
- Very fast
- Scriptable
- Automation friendly
- Handles large transfers efficiently

👉**AzCopy = Command Line + One-Way Sync**

2. Azure Storage Explorer ⭐⭐⭐
Azure Storage Explorer is a graphical desktop application for managing Azure Storage.
Supported on:
- Windows
- macOS
- Linux

Think of It As
File Explorer for Azure Storage

What Can It Do?
Upload Files
PC → Azure Storage

Download Files
Azure Storage → PC

Manage Storage
- Browse blobs
- Browse files
- Delete data
- Create containers
- Manage storage accounts

Interesting Fact
Storage Explorer actually uses:
AzCopy
behind the scenes for file transfers.

When to Use It?
Use when you prefer:
✅ GUI
instead of:
❌ Command Line

👉**Storage Explorer = GUI for Azure Storage**

3. Azure File Sync ⭐⭐⭐⭐
Azure File Sync lets you synchronize on-premises Windows Servers with Azure Files.
It creates a hybrid file storage environment.

Think of It As
Local File Server
↔
Azure Files
Both stay synchronized.

Synchronization Type
Important Exam Point
Azure File Sync provides:
Bi-Directional Sync
Local ↔ Azure

Main Benefits
Centralized Storage
Store master copy in Azure Files.

Local Performance
Frequently accessed files stay cached locally.

Cloud Backup

Azure acts as centralized storage.

Multiple Office Locations
Different offices can access the same synchronized files.

Supported Protocols
Azure File Sync supports:

- SMB
- NFS
- FTPS

Cloud Tiering ⭐⭐⭐
One of the most tested concepts.
What is Cloud Tiering?
Frequently used files:
Local Server
Rarely used files:
Azure Cloud

Benefit
Reduces local storage requirements.

Real-Life Example
Company has:
Delhi Office
Mumbai Office
Bangalore Office
All need access to the same file server.
Instead of maintaining separate servers:
Azure Files
↕
Azure File Sync
↕
Local Servers
All locations stay synchronized.

👉**Azure File Sync = Hybrid File Server Solution**

Quick Comparison Table ⭐⭐⭐⭐⭐

Feature	AzCopy	Storage Explorer	Azure File Sync
Interface	CLI	GUI	Sync Service
Upload/Download	Yes	Yes	Yes
Automation	Excellent	Limited	Automatic
One-Way Sync	Yes	Uses AzCopy	No
Bi-Directional Sync	No	No	Yes
Hybrid File Server	No	No	Yes
Best For	Bulk transfers	Easy management	Enterprise file sharing

If question mentions:
Command Line
Automation
Bulk Transfer
✅ AzCopy

If question mentions:
GUI
Visual Management
Browse Storage
✅ Azure Storage Explorer

If question mentions:
WindowsFileServer
HybridEnvironment
Two-Way Sync
Cloud Tiering
✅ Azure File Sync

Exam Questions
Q1. Which Azure tool provides a graphical interface for storage management?
✅ Azure Storage Explorer

Q2. Which Azure file transfer tool is command-line based?
✅ AzCopy

Q3. Is AzCopy synchronization one-way or two-way?
✅ One-way

Q4. Which service synchronizes Windows file servers with Azure Files?
✅ Azure File Sync

Q5. Which Azure service supports cloud tiering?
✅ Azure File Sync

Remember this:
AzCopy
= CLI + Fast Transfer + One-Way Sync
Storage Explorer
= GUI + Storage Management
Azure File Sync
= Hybrid File Server + Bi-Directional Sync + Cloud Tiering

Tool	Purpose
AzCopy	Command-line file transfer
Azure Storage Explorer	GUI-based file management
Azure File Sync	Hybrid file synchronization

Azure Identity, Access & Security:

Azure Directory Services:

Azure's identity and access management services revolve around:
Microsoft Entra ID (formerly Azure Active Directory)
It helps users securely sign in to:

- Azure
- Microsoft 365
- Cloud applications
- Custom applications

1. Microsoft Entra ID ⭐⭐⭐⭐⭐

Microsoft Entra ID is Microsoft's:
Identity and Access Management (**IAM**) service in the cloud.
It verifies:
✓ Who you are
✓ What you can access

Think of It As
Cloud Active Directory

Why Is It Important?
Every Azure environment uses identity management.
Before accessing:

- Virtual Machines
- Storage Accounts
- Databases
- Applications

users must first be authenticated.
Microsoft Entra ID handles this.

Who Uses Microsoft Entra ID?
IT Administrators
Manage:

- Users
- Groups
- Permissions
- Access Policies

Developers
Add:

- Login systems
- SSO
- Authentication

to applications.

End Users
Use it for:

- Sign-in
- Password reset
- Account management

Microsoft Cloud Customers
Used by:

- Microsoft 365
- Azure
- Dynamics 365

What Does Microsoft Entra ID Do?
1. Authentication ⭐⭐⭐
Verifies user identity before access is granted.
Features include:

- Password verification
- Multifactor Authentication (MFA)
- Smart Lockout
- Self-Service Password Reset (SSPR)

Example
Username + Password + OTP
This is authentication.

2. Single Sign-On (SSO) ⭐⭐⭐⭐
One login provides access to multiple applications.

Example
Login once:

```
Microsoft          Entra          ID
      ↓
Teams
Outlook
SharePoint
Azure Portal
```
No repeated logins.

Benefits
✓ Better user experience
✓ Fewer passwords
✓ Improved security

👉 **SSO = One Login, Many Apps**

3. Application Management
Manage:
- SaaS applications
- Enterprise applications
- Cloud apps

through a central location.
Examples:
- Salesforce
- ServiceNow
- Microsoft 365

4. Device Management
Works with:
`Microsoft Intune`
to manage devices.
Allows:
- Device registration
- Device compliance checks
- Conditional Access

Microsoft Entra ID Ecosystem
Main capabilities:
```
 Authentication
 Single                    Sign-On
 Application             Management
 Device Management
```

2. Microsoft Entra Connect ⭐⭐⭐⭐
Microsoft Entra Connect synchronizes identities between:
```
On-Premises     Active     Directory
          ↕
Microsoft Entra ID
```

Why Use It?
Without synchronization:
`Two separate user accounts`
must be maintained.
With Entra Connect:
`Single Identity`
across both environments.

Benefits
✓ User synchronization
✓ Password synchronization
✓ Hybrid identity
✓ Consistent login experience
Example
Employee account created in:
`On-Prem AD`
Automatically appears in:
`Microsoft Entra ID`

👉**Entra Connect = Hybrid Identity Bridge**

Synchronization Direction
```
On-Prem                          AD
      ↕
Entra                        Connect
      ↕
Entra ID
```
Synchronization occurs between on-premises AD and Entra ID.

3.**Microsoft Entra Domain Services**
⭐⭐⭐⭐
Provides managed domain services in Azure.
Supports:
- Domain Join
- Group Policy
- LDAP
- Kerberos
- NTLM

WITHOUT managing domain controllers.

Think of It As
`Active Directory as a Service`

Why Use It?
Some older applications require:
- LDAP
- Kerberos
- NTLM

and cannot use modern cloud authentication.

Entra Domain Services supports those applications.

Features
Domain Join
Join Azure VMs to a domain.

Group Policy
Apply organization policies.

LDAP Support
Supports directory lookups.

Kerberos / NTLM
Supports legacy authentication methods.

Major Advantage
You DO NOT manage:
Domain Controllers
Backups
Patching
Updates
Azure manages them.

👉 **Entra Domain Services = Managed Active Directory Features**

How It Works

```
On-Prem AD
      ↓
Entra Connect
      ↓
Microsoft Entra ID
      ↓
Microsoft Entra Domain Services
```

Important Exam Concept
Synchronization Direction
Entra Connect:
On-Prem AD ⇔ Entra ID
(Hybrid synchronization)

Entra Domain Services
Entra ID → Domain Services
(One-Way Sync)

Question ⭐⭐⭐⭐
Which service provides:
LDAP
Kerberos
NTLM
Group Policy
without deploying domain controllers?
✅ Microsoft Entra Domain Services

Comparison Table ⭐⭐⭐⭐⭐

Feature	Microsoft Entra ID	Entra Connect	Entra Domain Services
Identity Management	✅	❌	❌
Authentication	✅	❌	Limited
SSO	✅	❌	❌
MFA	✅	❌	❌
Hybrid Identity	❌	✅	❌
LDAP	❌	❌	✅
Kerberos	❌	❌	✅
NTLM	❌	❌	✅
Group Policy	❌	❌	✅
Managed Domain Controllers	✅ Azure-managed	N/A	✅ Azure-managed

If question mentions:
Authentication
MFA
SSO
User Identity
✅ Microsoft Entra ID

If question mentions:
Synchronize On-Prem AD
Hybrid Identity
✅ Microsoft Entra Connect

If question mentions:
LDAP
Kerberos
NTLM
Group Policy
Legacy Applications
✅ Microsoft Entra Domain Services

Exam Questions
Q1. What is the cloud-based identity service in Azure?
✅ Microsoft Entra ID

Q2. Which feature allows one login for multiple applications?
✅ Single Sign-On (SSO)

Q3. Which service synchronizes on-premises AD with Azure?
✅ Microsoft Entra Connect

Q4. Which Azure service provides LDAP and Kerberos support?
✅ Microsoft Entra Domain Services

Q5. Do you manage domain controllers in Entra Domain Services?
✅ No, Azure manages them.

Remember this hierarchy:

```
Microsoft Entra ID
      ↓
Identity & Authentication
```

```
Entra Connect
      ↓
Hybrid Identity Sync
```

```
Entra Domain Services
      ↓
LDAP + Kerberos + NTLM + Group Policy
```

```
Entra ID
= Who are you?

Entra Connect
= Sync identities

Entra Domain Services
= Legacy AD features in Azure
```

Azure Authentication methods:

Authentication verifies:
`Who are you?`
before Azure grants access.
Microsoft Entra ID supports several authentication methods:

1. Password Authentication
2. Single Sign-On (SSO)
3. Multifactor Authentication (MFA)
4. Passwordless Authentication

Authentication Methods Overview

Method	Security	Convenience
Password Only	Low	High
MFA	High	Medium
Passwordless	Highest	Highest

`Passwordless Authentication` offers both:
✅ High Security
✅ High Convenience

1. Password Authentication
What is it?
The traditional login method.
User provides:
`Username + Password`

Authentication Factor
Uses:
`Something You Know`
Example:

- Password
- PIN
- Security Question

Advantages
✓ Easy to use
✓ Familiar
✓ Fast login

Disadvantages
❌ Weakest security
❌ Password reuse
❌ Phishing attacks
❌ Password guessing

👉**Password Only = Convenient but Least Secure**

2. Single Sign-On (SSO) ⭐⭐⭐⭐

Single Sign-On allows:
One Login
↓
Many Applications
Example
User signs in once:
Microsoft Entra ID
Then automatically accesses:

- Outlook
- Teams
- SharePoint
- Azure Portal

without logging in again.

Benefits
Reduced Password Fatigue
One password instead of many.

Better User Experience
Fewer login prompts.

Easier Administration
Manage access from one identity.

Important Exam Point
SSO security depends on:
Initial Authentication
If the first login is compromised, all connected applications are affected.

👉 **SSO = One Identity, Many Applications**

3. Multifactor Authentication (MFA)
⭐⭐⭐⭐⭐
MFA requires:
More Than One Authentication Factor
before access is granted.

Why MFA?
Even if a password is stolen:
Password Alone
≠ Access

Authentication Factors
Factor 1: Something You Know
Examples:

- Password
- PIN
- Security Questions

Factor 2: Something You Have
Examples:

- Mobile phone
- Authenticator app
- Security token

Factor 3: Something You Are
Examples:

- Fingerprint
- Face recognition
- Retina scan

Example MFA Login
Password
+
Phone Notification
or
Password
+
Fingerprint

Microsoft Entra MFA
Microsoft provides built-in MFA through:
Microsoft Entra Multifactor Authentication

Common Verification Methods

- Authenticator App Notification
- SMS Code
- Phone Call
- Security Key

Benefits
✓ Prevents most credential attacks
✓ Stronger security
✓ Easy deployment

👉 **MFA = Two or More Factors**

4. Passwordless Authentication
⭐⭐⭐⭐⭐
Removes passwords completely.
Instead uses:
Trusted Device
+
PIN / Biometrics

Why Passwordless?
Passwords are:

❌ Forgotten
❌ Reused
❌ Stolen
❌ Phished
Passwordless removes these risks.

👉 **No Password = Nothing to Steal**

Passwordless Options in Microsoft Entra ID
Microsoft supports:

1. Windows Hello for Business
2. Microsoft Authenticator
3. FIDO2 Security Keys

4.1 Windows Hello for Business ⭐ ⭐ ⭐
Passwordless sign-in built into Windows devices.
Uses:

- Face recognition
- Fingerprint
- PIN

Best For
`Dedicated Work PCs`

Benefits
✓ SSO support
✓ Biometric security
✓ PKI integration

Example
```
Face Scan
   ↓
Access Windows + Azure
```

4.2 Microsoft Authenticator App ⭐ ⭐ ⭐
Turns a smartphone into an authentication device.
Available on:

- Android
- iPhone

Passwordless Login Flow
```
Login Request
   ↓
Phone Notification
   ↓
Approve
   ↓
Access Granted
```

Benefits
✓ No password
✓ Convenient
✓ Works anywhere

4.3 FIDO2 Security Keys ⭐ ⭐ ⭐ ⭐
What is FIDO2?
An open standard for passwordless authentication.
Uses physical hardware security keys.

Examples

- USB Key
- NFC Key
- Bluetooth Security Key

Login Process
```
Insert Security Key
   ↓
Authenticate
   ↓
Access Granted
```

Benefits
✓ Extremely secure
✓ Resistant to phishing
✓ No passwords stored

Best For

- Administrators
- High-security environments
- Shared workstations

👉 **FIDO2 = Physical Security Key**

Authentication Comparison ⭐ ⭐ ⭐ ⭐ ⭐

Method	Password Required	Security Level
Password Only	Yes	Low
SSO	Yes (once)	Depends on initial login
MFA	Yes + Extra Factor	High
Passwordless	No	Highest

If question mentions:
`One login for many apps`
✅ Single Sign-On (SSO)

If question mentions:
Password + Phone
Password + Fingerprint
✅ Multifactor Authentication (MFA)

If question mentions:
No Password
Biometrics
PIN
Security Key
✅ Passwordless Authentication

If question mentions:
USB Security Key
WebAuthn
Hardware Key
✅ FIDO2 Security Key

Exam Questions
Q1. What does SSO provide?
✅ One login for multiple applications

Q2. MFA uses how many factors?
✅ Two or more

Q3. Which factor category does a fingerprint belong to?
✅ Something You Are

Q4. Name three passwordless options in Microsoft Entra ID.
✅ Windows Hello for Business
✅ Microsoft Authenticator
✅ FIDO2 Security Keys

Q5. Which authentication method offers the highest security and convenience?
✅ Passwordless Authentication

Remember this progression:
Password
↓
MFA
↓
Passwordless
Security increases as you move right.

```
SSO
= One Login

MFA
= Multiple Factors

Passwordless
= No Password

FIDO2
= Physical Security Key
```

Azure External Identities:

Normally, Microsoft Entra ID manages users inside your organization.
But organizations often need to work with:

- Vendors
- Partners
- Suppliers
- Contractors
- Customers

Azure provides **Microsoft Entra External ID** to securely manage these external users.

What is an External Identity?
An external identity is:
A user outside your organization
who needs access to your resources.
Instead of creating and managing separate accounts, external users can:
Bring Their Own Identity
Examples:

- Work Account
- Microsoft Account
- Google Account
- Facebook Account
- Government-issued Digital Identity

Key Idea
ExternalProvider
↓
AuthenticatesUser

```
Your                                Tenant
      ↓
Authorizes Access
```

Why External Identities Matter
Organizations frequently collaborate with external people.
Examples:
Business Partner

```
Your                                Company
      ↓
Supplier
```

Supplier employees need access to shared resources.

Contractor
External contractors may need:

- Teams access
- SharePoint access
- Internal applications

Customer Portal
Customers need access to:

- Web applications
- SaaS platforms
- Self-service portals

Microsoft Entra External ID Capabilities
There are **3 major capabilities**:

1. B2B Collaboration
2. B2B Direct Connect
3. External ID for Customers (B2C)

1. B2B Collaboration ⭐⭐⭐⭐⭐
Business-to-Business (B2B) collaboration allows external users to access resources in your tenant.
They use:

```
Their Own Credentials
```

instead of receiving a new account.

Example

```
Supplier                          Employee
        ↓
Invited      to      Your      Tenant
        ↓
Access Teams / SharePoint
```

Directory Representation
External users appear as:

```
Guest Users
```

inside your directory.

Best For

- Vendors
- Contractors
- Partners
- Consultants

👉 **B2B Collaboration = Guest Users**

2. B2B Direct Connect ⭐⭐⭐⭐
Provides direct collaboration between:

```
Tenant A ⇔ Tenant B
```

using a trusted connection.

Main Use Case
Microsoft Teams Shared Channels
Users collaborate across organizations without becoming guest users.

Key Difference
B2B Collaboration

```
External                          User
     ↓
Guest Account Created
```

B2B Direct Connect

```
No          Guest          Account
Direct Tenant Trust
```

Benefits
✓ Easier collaboration
✓ No guest account management
✓ Better Teams integration

👉**B2B Direct Connect = Teams Shared Channels**

3. External ID for Customers (Formerly Azure AD B2C) ⭐⭐⭐⭐⭐
Designed for:

```
Customers
Consumers
Public Users
```

of your applications.

Example
Imagine you build:

- Shopping App

- Banking Portal
- Food Delivery App

Customers need to sign in.
External ID for Customers provides:
✓ Registration
✓ Login
✓ Identity Management

Customer Login Options
Users may sign in using:

- Google
- Facebook
- Microsoft Account
- Email Address
- Custom Identity Provider

Important
Unlike B2B users:
`Customers`
are managed in a:
`Separate External ID Tenant`

Best For
Customer-facing applications.

👉External ID for Customers = Customer Authentication

Comparison Table ⭐ ⭐ ⭐ ⭐ ⭐

Feature	B2B Collaboration	B2B Direct Connect	External ID for Customers
Target Users	Partners/Vendors	Other Organizations	Customers
Guest User Created	Yes	No	No
Uses Own Credentials	Yes	Yes	Yes
Teams Shared Channels	No	Yes	No
Customer Apps	No	No	Yes
Separate Tenant	No	No	Yes

Managing Guest Access ⭐ ⭐ ⭐
Microsoft Entra ID lets administrators:
Invite Users
`Partner → Guest User`

Review Access
Regularly verify whether users still need access.

Remove Access
Delete access when:

- Projects end
- Contractors leave
- Partnerships end

Access Reviews
Administrators can:
✓ Review permissions
✓ Recertify access
✓ Remove unnecessary access

Example
Scenario 1
A supplier needs access to SharePoint.
✅ Use:
`B2B Collaboration`

Scenario 2
Two companies need to collaborate through Teams Shared Channels.
✅ Use:
`B2B Direct Connect`

Scenario 3
An e-commerce website needs customer login.
✅ Use:
`External ID for Customers`
If question mentions:
`Guest User`
`Partner`
`Vendor`
`Contractor`
✅ B2B Collaboration

If question mentions:
`Teams Shared Channels`
`Cross-Tenant Trust`
✅ B2B Direct Connect

If question mentions:
`Customers`
`ConsumerApps`
`Public Users`
`Website Login`

✅ External ID for Customers (B2C)

Exam Questions

Q1. Which External ID feature creates guest users?

✅ B2B Collaboration

Q2. Which feature is designed for Teams shared channels?

✅ B2B Direct Connect

Q3. Which feature is used for customer-facing applications?

✅ External ID for Customers

Q4. Can B2B users use their own credentials?

✅ Yes

Q5. Who performs authentication in External ID scenarios?

✅ The external identity provider

```
B2B Collaboration
= Guest Users
```

```
B2B Direct Connect
= Teams Shared Channels
```

```
External ID for Customers
= Customer Login System
```

```
Partner → B2B Collaboration

Another Company → B2B Direct Connect

Customer → External ID for Customers
```

Azure Conditional Access:

Conditional Access is a Microsoft Entra ID feature that:

```
Evaluates Sign-In Conditions
            ↓
Makes Decision
            ↓
Enforces Access Policy
```

It determines whether a user should:

- Be allowed access
- Be required to perform MFA
- Be blocked completely

Think of Conditional Access as:

```
IF condition is met
THEN allow/block/require MFA
```

Example:

```
IF user logs in from office
→ Allow access

IF user logs in from unknown country
→ Require MFA

IF sign-in risk is high
→ Block access
```

Why Conditional Access?

Organizations want to:

✓ Protect sensitive resources

✓ Reduce unauthorized access

✓ Apply security only when needed

✓ Improve user experience

Instead of forcing MFA every time, Conditional Access can intelligently decide when to challenge users.

How Conditional Access Works

Step 1: Collect Signals

Microsoft Entra ID gathers information about the sign-in.

Step 2: Evaluate Policy

Rules are checked.

Step 3: Enforce Action

Access is allowed, restricted, or blocked.

Conditional Access Flow

```
Signals
   ↓
Decision
   ↓
Enforcement
```

Signals Used by Conditional Access

⭐⭐⭐⭐⭐

Conditional Access evaluates:

User or Group

Who is signing in?

Example:

```
Admins
Employees
Guests
```

Location

Where is the sign-in coming from?

Example:

```
Office Network
Home Network
Unknown Country
```

Device

What device is being used?

Example:

```
Managed Laptop
Personal Device
Mobile Phone
```

Application

Which application is being accessed?

Example:

```
Microsoft 365
Teams
Finance App
HR Portal
```

Sign-In Risk

Is the login suspicious?

Examples:

- Impossible travel
- Unusual location
- Compromised credentials
- Suspicious activity

Decisions Conditional Access Can Make

After evaluating signals, Azure can decide:

Allow Access

```
User looks safe
```

Result:

✅ Full access

Require MFA

```
User appears risky
```

Result:

✅ Must verify identity

Block Access

```
Sign-in highly suspicious
```

Result:

❌ Access denied

Enforcement Actions

The decision is enforced through policies.

Common actions:

Allow Full Access

Normal sign-in.

Grant Access With Conditions

Examples:

- Require MFA
- Require approved app
- Require managed device

Block Access

Completely deny sign-in.

Common Conditional Access Scenarios

⭐⭐⭐⭐⭐

1. Require MFA for Admins

Signal

```
Admin Role
```

Enforcement

```
Require MFA
```

Reason:

Administrators have elevated privileges.

2. Allow Approved Applications Only

Signal

```
Email Service Access
```

Enforcement

```
Allow Approved Client Apps
```

Example:

Only Outlook can access company email.

3. Require Managed Devices

Signal

```
Sensitive Application
```

Enforcement
Require Compliant Device
Example:
Finance app only accessible from company-managed laptops.

4. Block Untrusted Sources
Signal
Unknown IP
High-Risk Login
Enforcement
Block Access

Examples
Scenario 1
Employee logs in from office laptop.
Known User
Known Device
Known Location
Result:
✅ Allow access

Scenario 2
Admin logs in from home.
Result:
✅ Require MFA

Scenario 3
User logs in from another country unexpectedly.
Result:
✅ Require MFA
or
❌ Block access

Scenario 4
Compromised credentials detected.
Result:
❌ Block sign-in

Benefits of Conditional Access
Better Security
Access is granted only under safe conditions.

Reduced Risk
Blocks suspicious sign-ins automatically.

Flexible MFA
MFA only when needed.

Improved User Experience
Trusted users avoid unnecessary prompts.

Conditional Access vs MFA ⭐⭐⭐⭐

Feature	MFA	Conditional Access
Adds extra authentication factor	✅	Can require MFA
Always authentication-focused	✅	No
Uses policies and conditions	❌	✅
Can block access	❌	✅
Can require managed devices	❌	✅

```
MFA
= Additional Verification

Conditional Access
= Decision Engine
```

Conditional Access often decides:
Require MFA?
Yes or No

Exam Keywords ⭐⭐⭐⭐⭐
If you see:
Location-based access
Risk-based access
Require managed device
Require MFA for admins
Block risky sign-ins
👉 Answer is usually:
Azure Conditional Access

Exam Questions
Q1. What does Conditional Access evaluate first?
✅ Signals (user, location, device, app, risk)

Q2. Can Conditional Access require MFA?
✅ Yes

Q3. Can Conditional Access block sign-ins?
✅ Yes

Q4. What is a common use case?
✅ Require MFA for privileged/admin accounts

Q5. What feature restricts access to managed devices only?
✅ Conditional Access

```
Signals
(User + Device + Location + App + Risk)
            ↓
Conditional Access
            ↓
Allow / MFA / Block
 MFA = Extra Authentication

 Conditional Access = Smart Security Rules
```

```
 IF Admin
 → Require MFA

 IF Unknown Location
 → Require MFA or Block

 IF Unmanaged Device
 → Deny Access
```

Azure role-based access control:

Azure Role-Based Access Control **(RBAC)**
Azure RBAC is Azure's authorization system that controls:
`WHO can do WHAT ON WHICH RESOURCE`
It follows the **Principle of Least Privilege**:
Give users only the permissions they need and nothing more.

Simple Definition
Instead of assigning permissions one-by-one:
`Read`
`Write`
`Delete`
`Create`
`Manage`

Azure lets you assign a **Role**.
The role automatically contains a predefined set of permissions.

```
User
  ↓
Role
  ↓
Permissions
```

Why RBAC?
Without RBAC:
- Every permission must be assigned manually.
- Difficult to manage large teams.
- Higher chance of mistakes.

With RBAC:
✅ Easier administration
✅ Consistent permissions
✅ Better security
✅ Scalable management

Example
Suppose a Storage Account exists.
Developer only needs to view files.
Without RBAC:
`Assign Read`
`Assign View`
`Assign List`
Manually.
With RBAC:
`Assign Reader Role`
Done.

Key Components of RBAC
RBAC works using:
`Who + Role + Scope`
Who
User, Group, Service Principal, Managed Identity
Role
Permissions assigned
Scope
Where permissions apply

What is a Role?
A role is a collection of permissions.
Example:
`Reader`
`Contributor`
`Owner`

Common Built-in Roles ⭐ ⭐ ⭐ ⭐ ⭐

Reader

Can:

✅ View resources

Cannot:

❌ Create

❌ Modify

❌ Delete

Think:

`Read Only`

Contributor

Can:

✅ Read

✅ Create

✅ Modify

✅ Delete

Cannot:

❌ Assign permissions

Think:

`Manage Resources`

Owner

Can:

✅ Full access

✅ Read

✅ Create

✅ Modify

✅ Delete

✅ Assign RBAC permissions

Think:

`Complete Control`

RBAC Scope

Permissions are assigned at a specific scope.

Scopes:

```
Management Group
      ↓
Subscription
      ↓
Resource Group
      ↓
Resource
```

Scope Hierarchy ⭐ ⭐ ⭐ ⭐ ⭐

```
Management Group
      ↓
Subscription
      ↓
Resource Group
      ↓
Resource
```

From broadest to narrowest.

Scope 1: Management Group

Highest level.

Contains multiple subscriptions.

Example:

```
Company
├──      Production      Subscription
└─ Development Subscription
```

Useful for:

- Enterprise-wide governance
- Centralized administration

Scope 2: Subscription

Contains multiple resource groups.

Example:

`Production Subscription`

Permissions affect everything inside the subscription.

Scope 3: Resource Group

Contains related resources.

Example:

`RG-WebApp`

May contain:

- VM
- Storage Account
- Database

Scope 4: Resource

Smallest scope.

Example:

```
Single                                VM
Single                          Database
Single Storage Account
```

Inheritance in RBAC ⭐ ⭐ ⭐ ⭐ ⭐

Permissions flow downward.

`ManagementGroup`

↓
Subscription
↓
Resource Group
↓
Resource
This is called:
Inheritance

Example of Inheritance
Assign:
Owner
at
Management Group
Then user automatically gets Owner access to:

- All subscriptions
- All resource groups
- All resources underneath

Another Example
Assign:
Reader
at Subscription level.
User can view:
All Resource Groups
All Resources
inside that subscription.

How RBAC is Applied
RBAC applies to a:
Scope
Possible scopes:

- Management Group
- Subscription
- Resource Group
- Resource

RBAC Enforcement
RBAC is enforced through:
Azure Resource Manager (ARM)
Whenever a user performs an action:
Create VM
Delete Storage
Modify Database
Azure Resource Manager checks:
Does user have permission?
If yes:
✅ Allow
If no:
❌ Deny

Important Exam Point ⭐⭐⭐⭐⭐
RBAC controls:
Management Plane
Examples:

- Create VM
- Delete VM
- Configure Resource

RBAC does NOT directly control:
Data Plane
Examples:

- Reading database rows
- Accessing application data

Those permissions are managed separately.

RBAC Uses an Allow Model
Azure RBAC grants permissions.
Example:
Role 1:
Reader
Role 2:
Contributor
Assigned to same user.
Result:
Reader + Contributor permissions
Permissions are combined.

Example
Company Structure:

```
Management Group
    ↓
Production Subscription
    ↓
WebApp Resource Group
    ↓
VM + Storage + Database
```

IT Admin
Assigned:
Owner
at Subscription level.
Can manage everything.

Developer
Assigned:
Contributor
at Resource Group level.

Can manage application resources only.

Auditor
Assigned:
`Reader`
at Subscription level.
Can view everything but change nothing.

RBAC vs Conditional Access

RBAC	Conditional Access
Controls permissions	Controls sign-in conditions
Authorization	Authentication/Access decision
What can user do?	Should user be allowed in?
Owner, Reader, Contributor	MFA, Location, Device policies

Example

Conditional Access:
`Require MFA`
After successful login:
RBAC determines:
`Can user create VM?`
`Can user delete storage?`

```
Authentication
= Who are you?

Authorization
= What can you do?
```

Azure RBAC handles:
`Authorization`

Keywords ⭐⭐⭐⭐⭐

If you see:
`Owner`
`Reader`
`Contributor`
`Least privilege`
`Permission inheritance`
`Management Group`
`Subscription`
`Resource Group`
`Resource`
👉 Think:
`Azure RBAC`

Exam Questions

Q1. What does Azure RBAC control?
✅ Authorization to Azure resources

Q2. What are the 3 common built-in roles?
✅ Reader, Contributor, Owner

Q3. Which role provides full access and can assign permissions?
✅ Owner

Q4. What is the broadest RBAC scope?
✅ Management Group

Q5. Permissions assigned at a parent scope are inherited by child scopes?
✅ Yes

Q6. What principle does RBAC support?
✅ Principle of Least Privilege

Remember:
`RBAC=Who + Role + Scope`
`User + Reader / Contributor / Owner + Management Group / Subscription / Resource Group / Resource`

```
Authentication
→ Entra ID

Conditional Access
→ Decide Access

RBAC
→ Decide Permissions
```

Example:

```
Login Successful
       ↓
Conditional Access approves
       ↓
RBAC decides:
Can user create VM?
Can user delete Storage?
```

Zero Trust Model:

Zero Trust is a modern security approach based on:

```
Never Trust
Always Verify
```

Instead of assuming users or devices inside the company network are safe, Zero Trust assumes:

```
Breach is Possible
```

and verifies every access request.

Simple Definition

Traditional security:

```
Inside Network = Trusted
Outside Network = Untrusted
```

Zero Trust:

```
Trust Nobody Automatically
Verify Everything
```

Every request must be authenticated, authorized, and validated.

Why Zero Trust?

Modern environments include:

- Remote workers
- Cloud applications
- Mobile devices
- Personal devices (BYOD)
- Multiple locations

Traditional network boundaries are no longer enough.

Zero Trust provides security regardless of:

```
Location
Device
Network
```

Core Principle

Zero Trust assumes:

```
An attacker may already be inside the network
```

Therefore:

```
Verify every request
```

instead of trusting based on network location.

Three Pillars of Zero Trust ⭐⭐⭐⭐⭐

1. Verify Explicitly

Always authenticate and authorize using all available information.

Check:

- Identity
- Device
- Location
- Application
- Data sensitivity
- Risk signals

Example

```
User logs in
   ↓
Verify identity
Verify device
Verify location
Evaluate risk
```

Only then grant access.

2. Use Least Privilege Access

Give only the minimum permissions required.

Example

Developer needs:

```
Read Storage
```

Grant:

```
Read Only
```

Not:

```
Full Admin Access
```

Techniques include:

Just-In-Time (JIT)

Access granted only when needed.

Just-Enough-Access (JEA)

Only required permissions are provided.

3. Assume Breach

Operate as if attackers may already exist in the environment.

Actions:

- Limit attack spread
- Segment resources
- Encrypt data
- Monitor activity
- Detect threats quickly

Zero Trust Principles Summary

```
Verify                          Explicitly
      +
Least                           Privilege
      +
Assume Breach
```

Traditional Security vs Zero Trust

⭐⭐⭐⭐⭐

Traditional Perimeter Model

```
Request
   ↓
Inside                          Network?
   ↓
Trusted
   ↓
Full Access
```

Characteristics:

- Network-based trust
- VPN grants internal access
- Less verification after login

Zero Trust Model

```
Request
   ↓
Verify                          Identity
Verify                            Device
Verify                               Risk
Verify                          Location
   ↓
Grant Access
```

Characteristics:

- Identity-based
- Continuous verification
- Context-aware decisions
- Every request evaluated

Traditional vs Zero Trust Comparison

Traditional	Zero Trust
Trust internal network	Trust nobody automatically
Location-based trust	Identity-based trust
Verify once	Verify continuously
Focus on perimeter	Focus on identity and risk
Assume safe	Assume breach

What Does Zero Trust Check?

Identity

```
Who is requesting access?
```

Example:

- Employee
- Admin
- Guest

Device

```
Is the device trusted?
```

Example:

- Managed laptop
- Personal phone

Location

```
Where is the request coming from?
```

Example:

- Office
- Home
- Foreign country

Risk

```
Is the sign-in suspicious?
```

Example:

- Impossible travel
- Malware detected
- Unusual login behavior

Example in Practice

Suppose a user signs in from:

```
Public                          Wi-Fi
Personal Laptop
```

Traditional model might deny immediately because it's outside the network.

Zero Trust evaluates:

```
Identity
Device                          State
Location
Risk Level
```

Possible outcomes:

✅ Allow access

✅ Require MFA

❌ Block access

Relationship with Azure Security Services

Microsoft Entra ID

Provides:

- Authentication
- Identity verification

Multifactor Authentication (MFA)

Provides:

```
Verify Explicitly
```

Conditional Access
Provides:
Context-Based Decisions
Examples:

- Require MFA
- Block risky users
- Require managed devices

Azure RBAC
Provides:
Least Privilege Access

Example
Employee accesses Finance Application.
Step 1
Identity verified through:
Microsoft Entra ID
Step 2
Conditional Access checks:
Location
Device
Risk
Step 3
MFA required if needed.
Step 4
RBAC determines permissions.
Result:
Right User
Right Device
Right Access

Benefits of Zero Trust
Stronger Security
No automatic trust.

Better Protection
Limits damage if an account is compromised.

Supports Remote Work
Works anywhere.

Reduces Attack Surface
Least privilege minimizes exposure.

Faster Threat Detection
Continuous monitoring identifies suspicious activity.

Keywords ⭐⭐⭐⭐⭐
If you see:

Never trust, always verify
Verify every request
Least privilege
Assume breach
Identity-based security
Continuous validation
👉 Think:
Zero Trust Model

Exam Questions
Q1. What are the three principles of Zero Trust?
✅ Verify Explicitly
✅ Use Least Privilege Access
✅ Assume Breach

Q2. Does Zero Trust trust users inside the corporate network automatically?
❌ No

Q3. What does "Assume Breach" mean?
✅ Act as if attackers may already be inside the environment.

Q4. Which principle limits user permissions to only what is required?
✅ Least Privilege Access

Q5. Which Azure feature commonly implements risk-based access decisions?
✅ Conditional Access

```
Zero Trust
=
Never Trust
Always Verify
```

And:

```
Verify Explicitly
+
Least Privilege
+
Assume Breach
```

```
Traditional Security
=
Trust Network

Zero Trust
=
Trust Nothing
Verify Everything
```

Example:

```
Identity ✓
Device ✓
Location ✓
Risk ✓
       ↓
Access Granted
```

Defense in-depth:

Defense-in-Depth is a security strategy that uses **multiple layers of protection** instead of relying on a single security control.

Goal

```
Protect                              Data
↓
Prevent      Unauthorized      Access
↓
Reduce Impact of Attacks
```

The idea is:

```
If   one   security   layer   fails,
another layer is already protecting
the system.
```

Simple Definition

Think of a bank vault:

```
Security                             Guard
    ↓
Building        Access        Control
    ↓
Locked                               Room
    ↓
Vault                                Door
    ↓
Safe Deposit Box
```

An attacker must break through multiple layers.

Azure security follows the same approach.

Why Defense-in-Depth?

A single security control can fail.

Example:

```
Password Compromised
```

Without extra protection:

❌ Attacker gains access

With Defense-in-Depth:

```
Password
    ↓
MFA
    ↓
Conditional                          Access
    ↓
RBAC
    ↓
Encryption
```

Multiple barriers stop the attack.

Defense-in-Depth Layers ⭐⭐⭐⭐⭐

Azure organizes security into 7 layers:

```
Physical Security
        ↓
Identity & Access
        ↓
Perimeter
        ↓
Network
        ↓
Compute
        ↓
Application
        ↓
Data
```

Important

```
Data     is     at     the     center.
All other layers protect the data.
```

Layer 1: Physical Security

Purpose

Protect datacenter hardware and facilities.

Examples

- Security guards
- Cameras
- Biometric access
- Locked server rooms
- Badge access systems

Azure Example

Microsoft secures Azure datacenters physically.

```
Physical Security=Protect Hardware
```

Layer 2: Identity and Access ⭐ ⭐ ⭐ ⭐ ⭐

Purpose

Control who can access resources.

Technologies

- Microsoft Entra ID
- MFA
- SSO
- Conditional Access
- RBAC

Key Actions

```
Authenticate                    Users
Authorize                       Users
Audit Access
```

Example

User must:

```
Enter                          Password
↓
Complete                          MFA
↓
Access Granted
```

Exam Tip

```
Identity & Access=Who Can Access?
```

Layer 3: Perimeter

Purpose

Protect against external network attacks.

Common Controls

- Firewalls
- DDoS Protection
- Gateway Protection

Azure Examples

- Azure DDoS Protection
- Azure Firewall

Exam Tip

```
Perimeter=First Network Defense
```

Layer 4: Network

Purpose

Control communication between resources.

Key Concepts

- Network Segmentation
- NSGs
- Private Endpoints
- VPN
- ExpressRoute

Best Practices

✓ Deny by default

✓ Allow only required traffic

✓ Restrict inbound/outbound access

Example

```
Web                                Server
   ↓
Can              Talk               To
Database                           Server

Everything                          Else
   ↓
Blocked
```

Exam Tip

```
Network=Limit Communication
```

Layer 5: Compute

Purpose

Secure servers and compute resources.

Examples

- Virtual Machines
- Containers
- Azure Functions

Key Actions

- Apply patches
- Endpoint protection
- Malware protection
- Vulnerability management

Example

```
Keep            VM            Updated
Install                     Antivirus
Apply Security Patches
```

Exam Tip

```
Compute=Protect Servers & Workloads
```

Layer 6: Application

Purpose

Protect applications from vulnerabilities.

Focus Areas

- Secure coding
- Security testing
- Secret management
- Application hardening

Azure Example
`Azure Key Vault`
Used for:

- Secrets
- Keys
- Certificates

Best Practices
✓ Secure SDLC
✓ Vulnerability testing
✓ Store secrets securely

Exam Tip
`Application=Secure the Code`

Layer 7: Data ⭐⭐⭐⭐⭐
Purpose
Protect the actual data.
This is the most important layer because:
`Attackers Usually Want Data`

Key Controls

- Encryption at Rest
- Encryption in Transit
- Access Controls
- Backups
- Compliance Controls

Azure Services

- Azure Storage Encryption
- Azure Key Vault

Data Locations
Data may exist in:

- Databases
- Virtual Machines
- SaaS Applications
- Cloud Storage

Exam Tip
`Data=Ultimate Target`

Layer Summary Table ⭐⭐⭐⭐⭐

Layer	Purpose
Physical Security	Protect datacenter hardware
Identity & Access	Verify users and permissions
Perimeter	Stop external attacks
Network	Restrict communication
Compute	Secure VMs and workloads
Application	Secure code and apps
Data	Protect sensitive information

Real-World Example

Employee accesses a financial database.

Physical Layer

Datacenter secured.

Identity Layer

User signs in via Entra ID.

Perimeter Layer

Firewall filters traffic.

Network Layer

Only approved subnet can access database.

Compute Layer

Database VM patched.

Application Layer

App validates requests.

Data Layer

Database encrypted.

Defense-in-Depth vs Zero Trust

Defense-in-Depth	Zero Trust
Security architecture	Security philosophy
Multiple protection layers	Never trust, always verify
Focus on layered defenses	Focus on verification
Protects infrastructure	Protects access decisions

Relationship

```
Zero Trust
     +
Defense-in-Depth
     =
Strong Security
```

Keywords ⭐⭐⭐⭐⭐

If you see:

```
Multiple                    layers
Physical                    Security
Perimeter
Network                     Segmentation
Data                        Protection
Layered Security
```

👉 Think:

```
Defense-in-Depth
```

Exam Questions

Q1. What is the goal of Defense-in-Depth?

✅ Protect data using multiple security layers.

Q2. What is the outermost layer?

✅ Physical Security

Q3. What is the innermost layer?

✅ Data

Q4. Which layer uses MFA and SSO?

✅ Identity and Access

Q5. Which layer uses DDoS protection and firewalls?

✅ Perimeter

Q6. Which layer secures VMs and operating systems?

✅ Compute

Q7. Which Azure service commonly protects secrets and keys?

✅ Azure Key Vault

Encryption and key management in Azure:

Encryption in Azure

Encryption protects data by converting it into an unreadable format for unauthorized users.

```
Plain                               Data
    ↓
Encryption
    ↓
Cipher                              Text
    ↓
Only Authorized Users Can Read
```

Two Types of Encryption ⭐⭐⭐⭐⭐

For AZ-900, Microsoft heavily focuses on:

1. Encryption at Rest

Protects data when it is stored.

Examples:

- Azure Storage
- Azure SQL Database
- Managed Disks
- Databases

```
Stored                              Data
      ↓
Encrypted
```

Example

Customer records stored in Azure Storage remain encrypted even if someone gains physical access to the disks.

2. Encryption in Transit

Protects data while moving across networks.

Examples:

- User ↔ Website
- Application ↔ Database
- API ↔ API

```
User
  ↓                                 HTTPS/TLS
Application
  ↓                                 HTTPS/TLS
Database
```

Technologies

- TLS
- HTTPS
- VPN tunnels

Quick Comparison ⭐⭐⭐⭐⭐

Encryption Type	Protects
At Rest	Stored data
In Transit	Moving data

```
At Rest = Stored
In Transit = Moving
```

Practical Example
Suppose a banking application stores customer information.
Data Stored
`Azure SQL Database`
Protected by:
✅ Encryption at Rest

Data Sent to User
`Browser ↔ Web App`
Protected by:
✅ Encryption in Transit

Azure Key Vault ⭐⭐⭐⭐⭐
The MOST IMPORTANT service in this module.
Definition
Azure Key Vault is a service used to securely store and manage:
`Secrets`
`Keys`
`Certificates`

What Can Azure Key Vault Store?
1. Secrets
Examples:
- Passwords
- API Keys
- Connection Strings
- Tokens

`Database Password`
`API Key`
`Storage Key`

2. Encryption Keys
Used for:
- Encrypting data
- Decrypting data

Examples:
- RSA Keys
- AES Keys

3. Certificates
Examples:
- TLS Certificates
- SSL Certificates

Used to secure websites and applications.

Why Use Azure Key Vault?
Without Key Vault:
`Password`
`Stored in Source Code`
❌ Security Risk

With Key Vault:
`Application`
`↓`
`Azure Key Vault`
`↓`
`Secret Retrieved Securely`
✅ Secure

Benefits of Azure Key Vault ⭐⭐⭐⭐⭐
Access Control
Control who can view and use secrets.
Example:
`Developer → Read Only`
`Admin → Full Access`

Key Rotation
Rotate keys automatically.
`Old Key`
`↓`
`New Key`
Reduces security risk.

Auditing
Track:
- Who accessed a secret
- When it was accessed
- Which key was used

Useful for compliance.

Why Key Management Matters
Good key management helps:
Security
Protect sensitive information.
Compliance
Meet regulatory requirements.
Auditing

Track key usage.
Separation of Duties
Different people can:
View
Use
Rotate
keys separately.

Managed Identities + Key Vault ⭐ ⭐ ⭐
Applications can securely retrieve secrets from Key Vault.
Instead of:
Password Hardcoded
Use:
Application
↓
Managed Identity
↓
Key Vault
This is more secure and commonly recommended by Microsoft.

Azure Key Vault Summary
Azure Key Vault
↓
Secrets
Keys
Certificates
Remember:
Key Vault = Secure Storage for Sensitive Information

Exam Keywords ⭐ ⭐ ⭐ ⭐ ⭐
If you see:
Store passwords securely
Store API keys
Store connection strings
Manage certificates
Rotate encryption keys
👉 Answer:
Azure Key Vault

Exam Questions
Q1. What protects stored data?
✅ Encryption at Rest

Q2. What protects data moving across networks?
✅ Encryption in Transit

Q3. Which Azure service stores secrets, keys, and certificates?
✅ Azure Key Vault

Q4. What are the three items stored in Key Vault?
✅ Secrets
✅ Keys
✅ Certificates

Q5. Which protocol commonly provides encryption in transit?
✅ TLS/HTTPS

Q6. Should passwords be stored directly in application code?
❌ No
Use:
✅ Azure Key Vault

```
Encryption at Rest
= Stored Data Protection

Encryption in Transit
= Moving Data Protection

Azure Key Vault
= Secrets + Keys + Certificates
```

Most Tested Facts
⭐ Encryption at Rest = Storage, Disks, Databases
⭐ Encryption in Transit = HTTPS/TLS
⭐ Azure Key Vault stores:
- Secrets
- Keys
- Certificates

⭐ Key Vault is Microsoft's primary service for secure secret and key management

```
Data Stored?
→ Encryption at Rest

Data Moving?
→ Encryption in Transit

Password/API Key/Certificate?
→ Azure Key Vault
```

Microsoft defender for cloud:

Cloud Security Posture Management (CSPM)
+
Threat Protection Service
It helps you:
✅ Assess security posture
✅ Secure resources
✅ Detect and respond to threats

Simple Definition for Exam
If the question says:
"A service that continuously assesses Azure resources, provides security recommendations, and generates security alerts"
✅ Answer:
Microsoft Defender for Cloud

Three Main Functions ⭐⭐⭐⭐⭐
Remember:
Assess
Secure
Defend

1. Assess
Continuously checks your environment for vulnerabilities.
Examples:
- Vulnerable VMs
- Unpatched servers
- Misconfigured storage accounts
- SQL security issues

Provides:
✅ Security recommendations
✅ Vulnerability assessments

2. Secure
Uses best-practice security recommendations.
Works with:
Azure Policy
Provides:
- Security controls
- Secure Score
- Compliance recommendations

3. Defend
Detects active threats.
Provides:
- Security alerts
- Threat detection
- Attack investigation
- Incident response guidance

Secure Score ⭐⭐⭐⭐⭐
Very common AZ-900 question.
What is Secure Score?
A measurement of your Azure security posture.
Higher Score = More Secure Environment
Defender for Cloud calculates:
- Current security state
- Missing protections
- Recommended improvements

Security Alerts ⭐⭐⭐⭐
When threats are detected, Defender for Cloud creates alerts.
Alerts include:
- Affected resource
- Threat details
- Recommended actions

Example:
Suspicious VM login attempt
Defender for Cloud:
✅ Detects
✅ Alerts
✅ Suggests remediation

Azure-Native Protection ⭐⭐⭐⭐
Protects Azure services such as:
Compute
- Virtual Machines

- AKS Containers

Databases
- Azure SQL

Storage
- Azure Storage Accounts

Applications
- Azure App Service

Networks
- Network resources

Just-In-Time (JIT) VM Access ⭐⭐⭐⭐

Frequently tested.

Problem

VM ports like:

```
22 (SSH)
3389 (RDP)
```

remain open continuously.

Attackers may exploit them.

Solution

Defender for Cloud JIT:

```
Ports Closed By Default
↓
Open Temporarily
↓
Automatically Close
```

Benefits:

✅ Reduced attack surface

✅ Better VM security

Hybrid and Multi-Cloud Support ⭐⭐⭐⭐

Defender for Cloud protects:

Azure

Native support

On-Premises Servers

Using:

```
Azure Arc
```

AWS

Supported

Google Cloud (GCP)

Supported

Exam Question

Which Azure service helps secure AWS and GCP resources from a centralized security dashboard?

✅ Microsoft Defender for Cloud

Azure Arc Connection ⭐⭐⭐⭐

To protect non-Azure resources:

```
On-Premises
↓
Azure Arc
↓
Defender for Cloud
```

Remember:

Azure Arc extends Azure management and security to non-Azure resources.

Vulnerability Assessment ⭐⭐⭐⭐

Defender for Cloud can assess:
- VMs
- Container Registries
- SQL Servers
- Databases

Purpose:

```
Find Security Weaknesses Before Attackers Do
```

Advanced Threat Protection ⭐⭐⭐⭐

Provides protection for:
- VMs
- Databases
- Containers
- Applications
- Networks

Detects:
- Malware
- Suspicious logins
- Brute-force attacks
- Exploits

Quick Comparison